



LEAP Project Team Cambodia

Energy Systems for Sustainable Development MJ2508

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- Export final of results from LEAP model and collaboration in results analysis (Demand, Installed capacity, Electricity generation and reserve margin) as collaboration in graphs Analysis building
- Collaboration and coordination during the building of final document and

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- Presentation Slides

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- Writing and review of section 3.4.8, 4.1.2, 4.1.4, 4.1.5, 4.2.2, 4.3.3

1. Executive Summary

This study seeks to enhance comprehension of Cambodia's electricity and energy sectors. The focus of the report is on assessing the interrelation of Energy-Environment-Economy, specifically examining Electricity-CO₂-Cost. Two distinct scenarios are evaluated: the Nationally Determined Contributions (NDCs) aligned with Cambodia's commitments under the Paris Agreement, targeting a global warming limit of under 2°C, and the Own Vision Scenario, a hypothetical projection by the LEAP Project Team Cambodia. The latter aspires to achieve universal electricity access by 2030 and a complete phase-out of fossil fuels in the electricity mix by 2050.

A detailed overview of the Low Emissions Analysis Platform (LEAP) and its scenario analysis functionality is presented in the report, emphasizing its significance in evaluating Cambodia's electricity usage. The report encompasses the model setup, key assumptions, and data collection methodologies for demand projections, installed capacity, and costs.

LEAP is utilized to generate demand projections, installed capacity, and cost estimates for each scenario. The results and discussion section provides an in-depth analysis of demand, installed capacity, electricity generation, and reserve margin for each scenario.

Addressing the research questions posed in the report, it explores the potential for replacing planned fossil fuel capacities with renewable energy sources. Furthermore, it evaluates whether the NDC and government policies align with sustainable development goals. The report concludes by offering valuable insights into the viability and potential of renewable energy in Cambodia.

2. Background and Literature Review

2.1. Cambodia's Background

The country of Cambodia, located in Southeast Asia, is a vibrant and culturally rich country with a population of 16.7 million and land area of 181,035 km² (BBC News, 2022). The population density per km² has slightly grown from 82 to 95 from 2013 to 2022, respectively.

The country's GDP is mainly driven by the textile and tourism industry as well as agriculture, although climate change has had a significant impact on the share. The average growth in GDP has been 7.1% over the past decade, from 61,525 billion riels (15,191 million USD) in 2013 to 144,170 billion riels (35,562 million USD) in 2022. The export-oriented manufacturing as well as the regrowth of tourism has dampened the economic effects of the Covid-19 pandemic, allowing for steady growth as per the figure below (Planning, 2019).

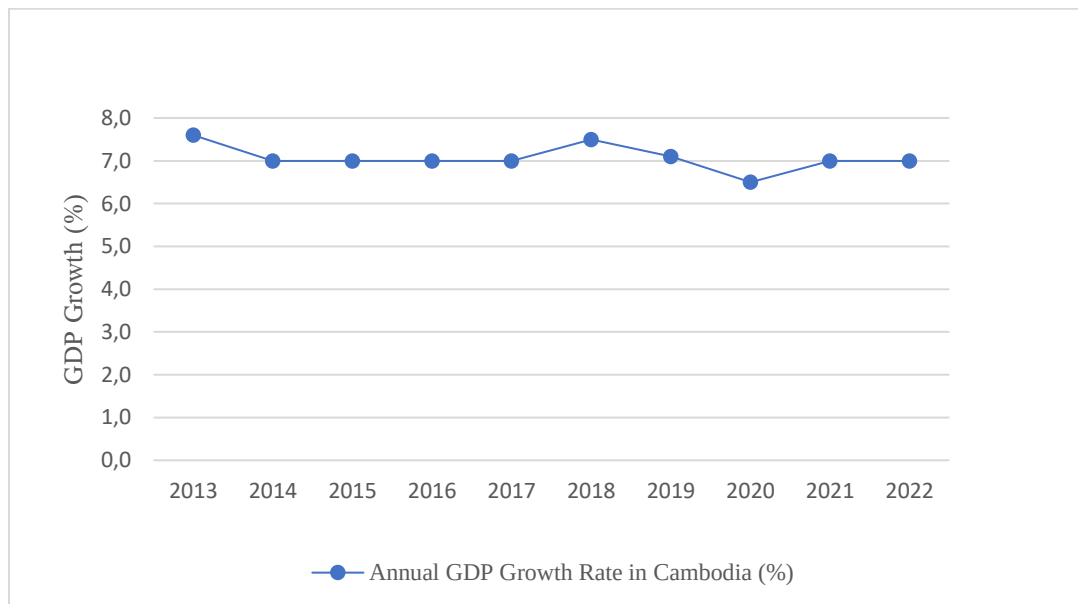


Figure 1: Annual GDP Growth Rate in Cambodia

The rapid economic growth, urbanization, and proposed policies in Cambodia has unavoidably led to a drastic increase in electricity consumption, and thus the GHG emissions. In terms of the annual CO₂ emissions, the country has been experiencing a sharp increase in the past decade. Figure 2 indicates a CO₂ production of 19.96 million tons CO₂ for the year 2022 (Roser, 2022), almost triple the amount of just two decades ago. This can be traced back to factors such as the steady rise

in industrial demand and complete electrification of the household sector. However, most of the country's GHG production lies in Forestry and Land Use (FOLU) at 49.2%, followed by the energy sector (22.2%), agriculture (17.5%) and industry (9%) among other contributors (López, 2019). Additionally, it is worth mentioning that the energy consumption per capita has risen from 1,591 kWh in 2010 to 4,035 kWh in 2021 (Roser, 2022). Although nowhere near the EU average of 6.07 MWh per capita (Anon., 2022), it can nevertheless be accounted for as a significant contributor.

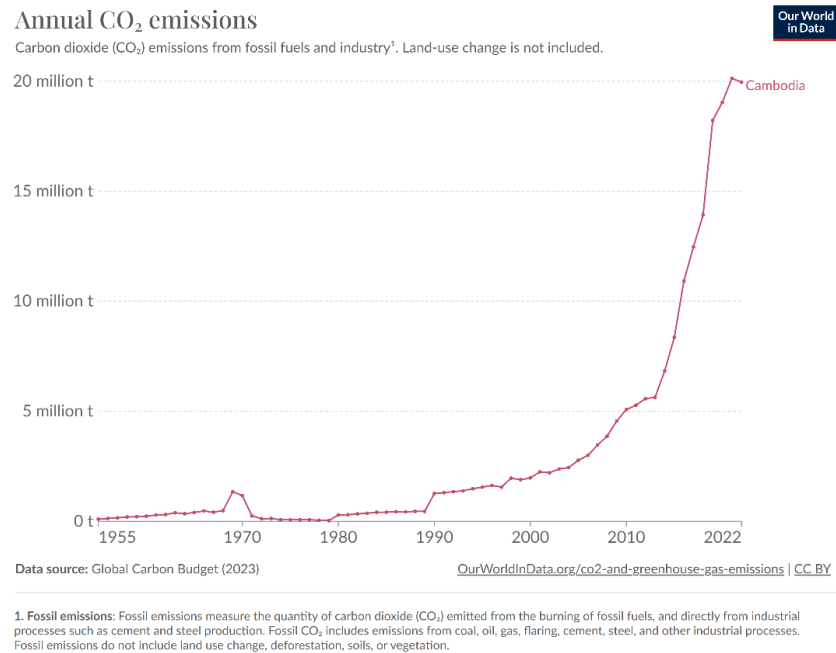


Figure 2: CO₂ Production in Cambodia from 1955 to 2022

As of 2023, the SDG Index of Cambodia is 103 out of 166 (Report, 2022). This index evaluates the country's progress in achieving the United Nations Sustainable Development Goals (SDGs). Cambodia's SDG rank indicates that the country is lagging and that there is room for improvement and effort in the economic, social and environmental aspects of the SDGs. According to the 2023 Sustainable Development Report, regarding SDG7, access to electricity has improved from 31.1% in 2010 to 86.4% in 2020. Although remarkable, the figures are proportional to the rise in GHG emissions as investments in industry and conventional power generation rise as well as the targets outlined in NDC not being on track.

Regarding SDG13, Cambodia has promising results. According to the Sustainable Development Report, Cambodia is not far off from reaching the objectives for SDG13 since the emissions from

cement production and fossil fuel imports and exports remain low. This can be because cement production is not within Cambodia's industrial scope and the imports and exports to neighboring countries remain minimal.

According to the NDC published by the government of Cambodia in 2020, most of the proposed targets will be in the agriculture, infrastructure (land and planning), and water resources. This plan outlines the fact that most of the reforms towards accomplishing SDGs are not projected onto the consumers (demand side), but rather more fundamental reform and investment from key players in ministries and industries. Key governmental actors such as Ministry of Agriculture, Forestry and Fisheries (MAFF), Ministry of Land Management, Urban Planning and Construction (MLMUPC), and Ministry of Public Works and Transport (MPWT) have announced serious mitigation and adaptation measures within the country's NDC, underscoring a serious intention for change. However, some targets remain ambitious and need further investigation for defining a realistic trajectory for change.

2.2. Cambodia's Energy and Electricity Balances

Despite falling outside the scope of our model, a thorough understanding of Cambodia's energy mix in terms of origin and end-use helps us better understand the country's present electricity mix, which is a significant part of our model, the situation, and how they are evolving to meet the SDG targets. So, fossil fuels and biofuels dominate the energy mix of Cambodia. Of the 95400 GWh of the energy supply in 2020, 42% were from biofuels and waste, 40% were from oil, and 13.5% were from coal sources, leaving only around 4.5% for renewable sources of energy, 4.2% for hydro and 0.3% for solar, wind and other renewable sources (Figure 3).

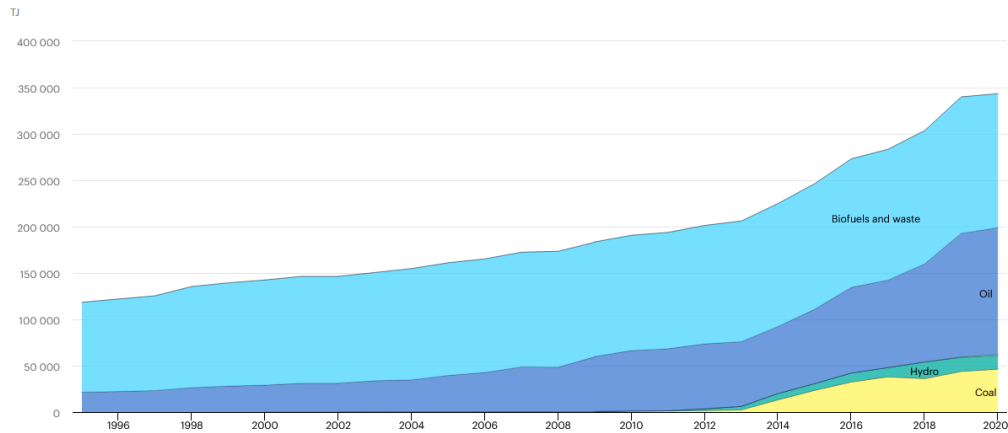


Figure 3: Cambodia's Total Energy Supply by Source 1995-2020. (IEA, 2023)

The Total Final Consumption (TFC) in Cambodia can be studied by sector. In 2020, 35.9% of TFC was consumed by the residential sector, 31.1% by industry, 23.9% by transport and 6.1% by the commercial sector. By source, 36563GWh of biofuels and waste, 34237GWh of oil products, 11092 GWh of electricity, and 1025GWh of coal are consumed annually in Cambodia. In the following Sankey Diagram (Figure 4) the Energy Balance of Cambodia can be visualized, we can see that most of the coal and oil supply comes from imports, while Biofuels and Waste is produced inland.

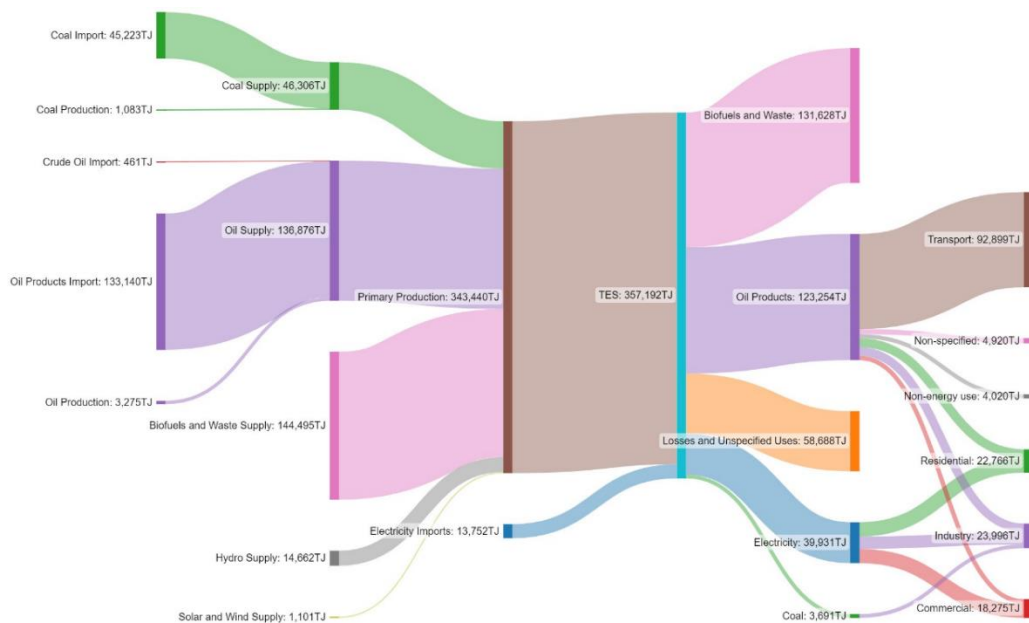


Figure 4: Energy Sankey Diagram for Cambodia in 2020. (IEA, 2023)

For the Electricity Balance of Cambodia, as of 2020, the relevant values and information for our model, much of the electricity comes from Coal and Hydro Power, while a smaller portion comes from Oil, Solar PV, and Biofuels. 4135GWh comes from Coal, 4073GWh from Hydro, 400GWh from Oil, 306GWh from Solar PV and 69GWh from Biofuels (Shares in Figure 5). The final percentage of Renewable penetration in electricity generation of 49.1%. There is also a percentage that comes from Imports, 3820GWh. This Electricity Supply is then used in 3 different sectors as of 2020, 3342GWh for Industry, 3881 GWh for Residential, and 3868GWh for Commercial. These values present our Current Accounts for Electricity Demand in our model and can be visualized in Figure 6. (IEA, 2023)

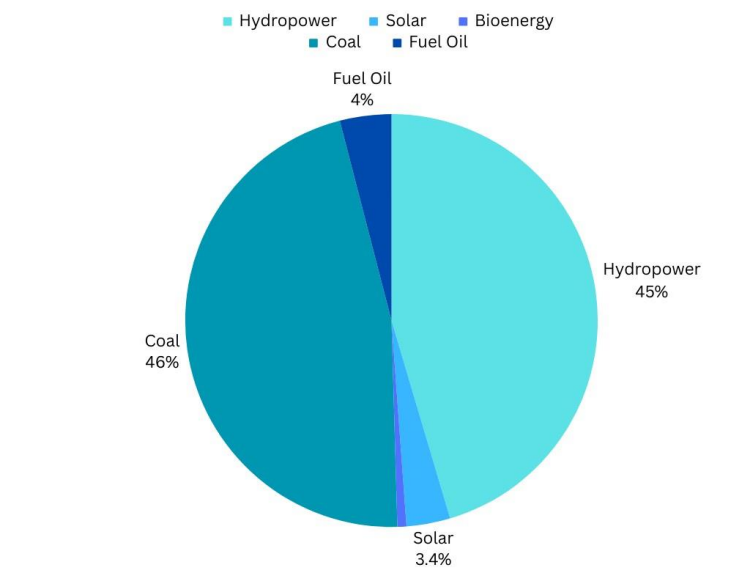


Figure 5: Electricity Mix by Energy Source in Cambodia. (IEA, 2023)

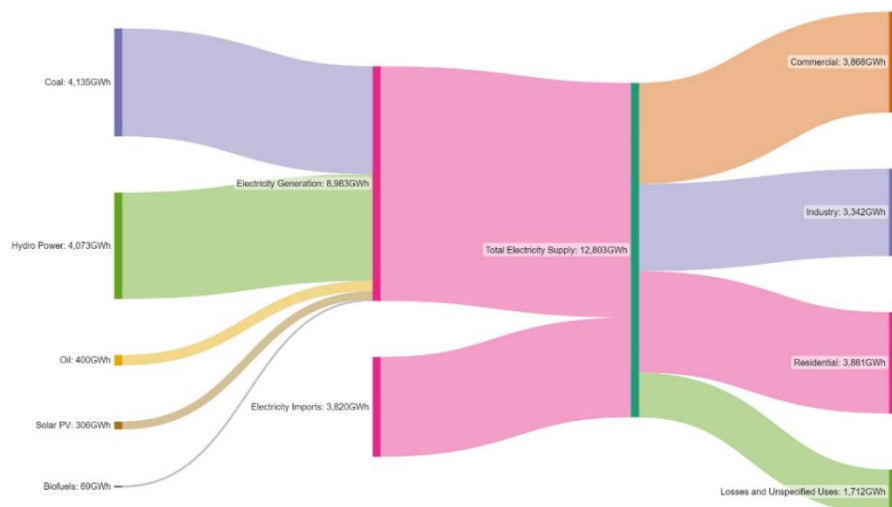


Figure 6: Electricity Sankey Diagram for Cambodia in 2020. (IEA, 2023)

2.3. Current and future operating technologies in the electricity sector

The current operating technologies in the electricity and energy sectors can be seen in the following charts. In the electricity sector we can see that hydropower and Coal is most of the installed capacity, while Biofuels and Oil are the majority in the Power Sector.

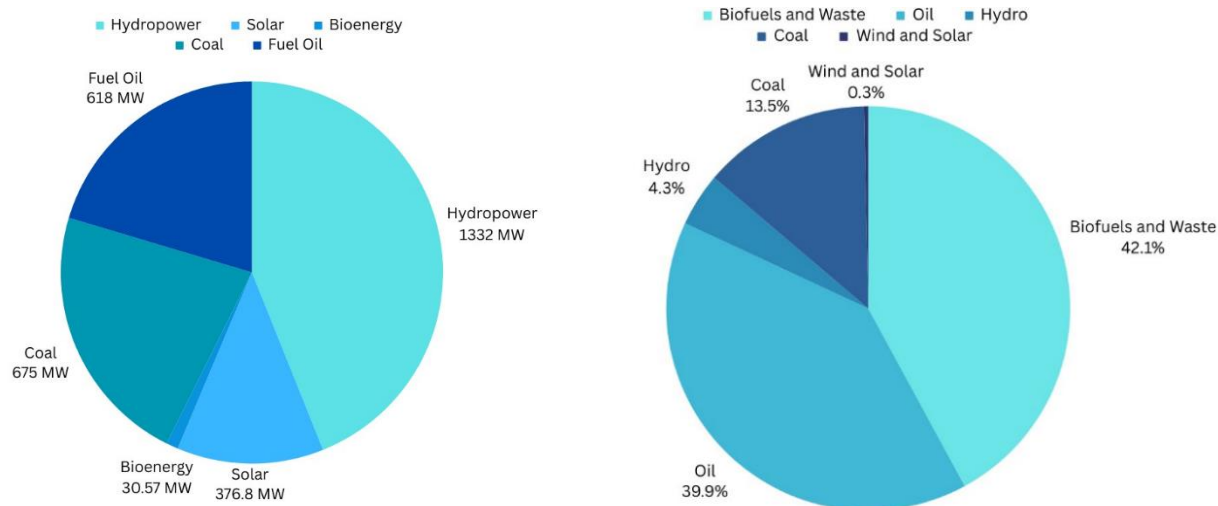


Figure 8: Installed Capacity of the Electricity Sector in 2021. Figure 7 : Installed Capacity of the Energy Sector in 2021. (IEA, 2023)

The planned installed capacities and imports in the electricity sector for 2040 stated in the “Power Development Masterplan” are presented in Table 1 together with the already installed capacities in 2021 and their progress in 2030.

Table 1: Installed and Planed Capacities and Imports for the Electricity Sector by Energy Source. (IEA, 2023) (Ministry of Mines and Energy, 2022)

Source	Power Source	2021	2030	2040
Domestic Generation	Hydropower	1332	2973	2973
	Solar	376,8	1005	3155
	Bioenergy	30,57	98	198
	Coal	675	2266	1591
	Fuel Oil	618	490	490
	Natural Gas	-	-	900
	Total Domestic Generation	3032,37	5417	9982
Imports	Laos	371	3095	3095
	Thailand	277,3	700	1000
	Vietnam	332,45		
	Total Imports	980,75	3795	4095
Total Capacity [MW]		4013,12	9212	14077

2.4. Policies and Mitigation Plans

Cambodia's Governmental Authorities have put forward multiple mitigation and strategy plans for the energy sector in the country. The Ministry of Environment is responsible for “Long-Term Strategy for Carbon Neutrality” (Ministry of Environment, 2021), “Cambodia's Updated Nationally Determined Contribution” (Ministry of Environment, 2020) and “Cambodia's National Cooling Action Plan” (Ministry of Environment , 2022), while the “Power Development Masterplan” (Ministry of Mines and Energy, 2022) and “National Energy Efficiency Policy” (National Committee on Energy Efficiency, 2022) were conducted by the Ministry of Mines and Energy.

Given Cambodia's vulnerability to climate change, these plans address diverse sectors, including Energy and Electricity - of particular interest in our project - as well as Agriculture, Forestry, Industry, Industrial Processes, Product Use, Waste, Tourism, Education, Health, Infrastructure, Biodiversity, and Water Resources. These policies significantly impact almost all Sustainable Development Goals (SDGs), as detailed in the National Determined Contributions (NDC) analysis of each policy.

The NDC outlines varied energy policies ranging from implementing building codes and efficiency labels (impacting SDGs 5, 7, 8, 9, 11) to achieving total electrification in rural areas (SDGs 1, 10). Noteworthy proposals include increasing the share of renewable energy resources, diversifying household energy sources, promoting off-grid street lighting, enhancing cooling efficiency in public sector buildings, and encouraging sustainable energy practices in manufacturing.

From “Long-Term Strategy for Carbon Neutrality” the mitigation actions of interest to the electricity sector for the baseline scenario were:

- Moderate penetration of electric vehicles - 70% for motorcycles and 40% for cars and urban buses by 2050;
- Fuel switching to electricity for cooking. (Ministry of Environment, 2021)

From “Cambodia’s Updated Nationally Determined Contribution”, the policies and mitigation policies/activities of use are:

- Application of electrical equipment’s labelling & MEPS (Lighting, Cooling and Equipment) which results in a reduction of 1.2TWh (29.7%) of electricity use in 2030 both in the Building Residential and Building Commercial sector;
- Building codes and enforcement/certification for new buildings and those undergoing major renovation resulting in a 10% reduction of electricity consumption in 2030 in the Building Commercial sector;
- Introduction of efficient electrical motors and boilers, leading to a 2.3% reduction of electricity consumption in 2030 in the Building Commercial and Industrial sectors. (Ministry of Environment, 2020)

“Cambodia’s National Cooling Action Plan” searches to improve conventional cooling by increasing the efficiency of air conditioning and refrigeration equipment and demand response measures. (Ministry of Environment , 2022)

The “Power Development Masterplan” presents the expansion plans for the power generation in Cambodia seen in Table 1 and Section 3.4.5. (Ministry of Mines and Energy, 2022)

The National Energy Efficiency Policy proposes to increase the efficient use of energy in Cambodia by targeting 17% savings in the electricity supply for the residential sector by 2030.

2.5. Aim of study/research question

With this project the intention to better understand the operation of the electricity and energy sectors of Cambodia, while doing a 3-E analysis, evaluating the correlation of Energy-Environment-Economy with a focus on Electricity-CO₂-Cost. This analysis will better aid in understanding the effects of the NDC and other government policies in the sector, and the effect of our own scenarios and its policies.

The main research questions we hope to answer are:

- To what extent can the planned fossil fuel capacities be replaced with renewable energy sources?
- Are the NDC and government policies in line with sustainable development?

3. Methodology

3.1. Description of LEAP

The Low Emissions Analysis Platform, or LEAP, is an adaptable software program made for assessments of climate change mitigation and energy policy analysis. LEAP is a tool for building models of various energy systems, each with unique data structures. It supports a wide range of modeling approaches, including different supply-side accounting, simulation, and optimization techniques as well as top-down macroeconomic modeling and bottom-up end-use accounting techniques. The scenario analysis functionality of LEAP is essential when evaluating Cambodia's electricity use. By contrasting energy needs, socioeconomic costs and benefits, and environmental effects, analysts can develop and assess different scenarios. The Scenario Manager on the platform makes it possible to describe specific policy measures.

3.2. Model Set-Up

Cambodia, as seen, is a developing country and with it comes the lack of some crucial data for the development of this model. As a result, the model created in LEAP is not as detailed as some of the examples provided.

It starts with the Key Assumptions, which are going to be further presented in the following section. Then, for the demand, the model was divided into Residential, Commercial, Industry and Transport. For Residential, only electrified houses are considered, and for these the electricity consumption is considered and broken-down into Comfort, Cooking, Recreation, Lighting and Cleaning Use. Commercial and Industry, due to lack of more information about their various consumers are only considered through its total electricity demand. Transport demand is divided into Cars and Motorcycles.

Following the demand, the Electricity Load Shape is defined, its explanation will be provided later in the report. Then, Transformation is split up into Transmission and Distribution, and Electricity Generation. The first one leads into Processes, where the only one considered is Electricity. The second one is divided into the Output Fuels which again is Electricity, and then into Processes and here the existing and planned electricity transformation technologies are specified:

- Existing Coal steam
- Existing Hydro
- Existing Oil powered plant
- Existing Solar
- Existing Biomass
- Planned Natural Gas
- New Solar
- New Hydro
- New Wind
- New Biomass

3.3. Key Assumptions

The Key Assumptions considered for Cambodia and its LEAP model were Income, Population, Household size, Households, GDP, Population Growth Rate, GDP Growth Rate and Electricity Prices (staff, 2022). Their values are presented in Table 2:

Table 2: Key Assumptions. (staff, 2022)

Population	16.6 Million People
Household Size	4.3 People per Household
Households	3.9 Million Households
GDP	27.0 Billion USD
Population Growth Rate	1.2%
GDP Growth Rate	3%
Electricity Price	0.012 (\$/kWh)

3.4. Data Collection

3.4.1 Demand Households (Current Accounts)

In 2021, total electricity consumption in Cambodia is 12.2 TWh, in which the residential sector accounts for 3881.12 GWh (IEA, 2023). The percentage of electricity consumption by types of appliances is derived from a 2020 study (Noeurn, 2021), applying the percentage of electricity consumption by types of appliances, which can be seen in Figure 9, and considering number of households, the total and per household electricity consumption by types of appliances is obtained as Table 3.

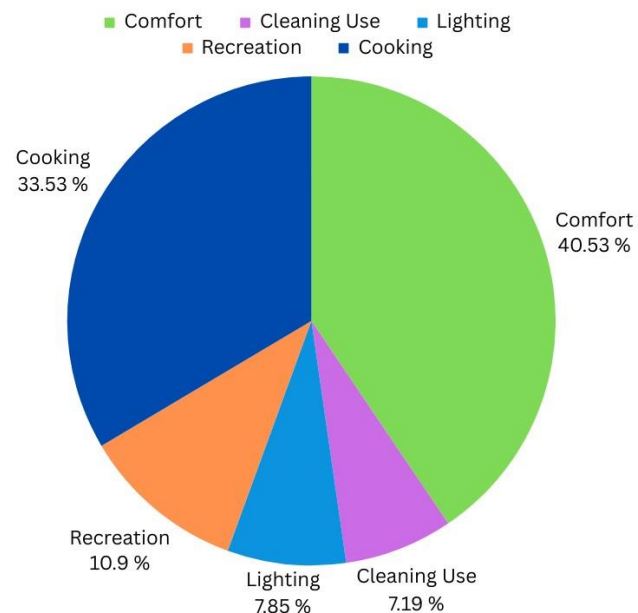


Figure 9: Breakdown of electricity use by end-use in the Household sector. (Noeurn, 2021)

Table 3: Total and per household electricity consumption by types of appliances. (Noeurn, 2021)

Appliances categories	Share	Electricity (GWh)	Electricity (kWh/H)
Comfort	41%	1573.0	403.3
Cooking	34%	1301.3	333.7
Recreation	11%	423.0	108.5
Lighting	8%	304.7	78.1
Cleaning-use	7%	279.1	71.6
Total	100%	3881.12	1165.5

3.4.2 Demand Commercial (Current Accounts)

The Commercial Sector of Cambodia is one of the most electricity consuming sectors, just below Residential and due to the current Electricity mix, a large amount of global CO₂ emissions come with it. These come from various sources and types of buildings, Office, Retail, Hotel, Healthcare, Education, Warehouses and Public Institutions that have reached as of 2021 21.56 million square meters of area. As with the other sectors, and with the increase of the country's GDP and Population, so will the demand for these commercial buildings increase. But as of 2020, the Commercial Sector consumed 3868 GWh of electricity during the year, 36.75% of total consumption. (IEA, 2023) (National Committee on Energy Efficiency, 2022) (Dul, 2020)

3.4.3 Demand Industry (Current Accounts)

Cambodia's Industry sector is a significant contributor to the country's global CO₂ emissions. Most of the revenue from the industrial sector can be traced back to agriculture and exports, textiles, and manufacturing. According to Cambodia's latest energy consumption statistics presented by the International Energy Agency (IEA), the total electricity consumption of Cambodia in 2020 was evaluated to be 40,000 TJ, which equates to 11.11 TWh as shown in Figure 10. (IEA, 2020)

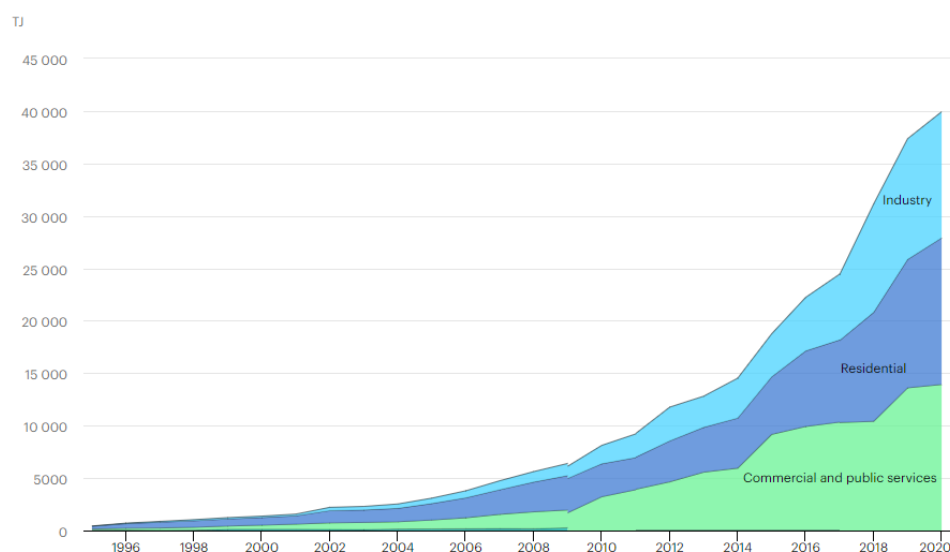


Figure 10: Share of electricity use by sector. (IEA, 2023)

As a part of the overall sum, industry comprises 30.01% of it with a capacity of 12,031 TJ or 3341.94 GWh. Given the strong targets for strategy and investment in this sector, the demand for industry is projected to grow.

3.4.4 Demand Transport (Current Accounts)

In Cambodia's energy mix, the transportation sector accounts for the lion's share of end-use. Because most cars are fuel-powered, the bulk of their energy comes from oil products, and oil is one of the country's biggest sources of GHG emissions. As a result, transitioning from gasoline vehicles to EVs and electric motorbikes is a critical step in lowering the country's GHG emissions, and both the NDC and “Long-Term Strategy for Carbon Neutrality” refer to the country’s efforts to increase the share of these vehicles in the roads in the upcoming years, which makes it a necessity in our model since it is a growing sector in the Electricity Mix.

The main mode of transport in Cambodia remains to be road transport, with 61,534 km of road length (ESCAP, 2023). Cambodia’s enhancement of rural connectivity has been a discussion for years, and multiple policy frameworks have promised road development and infrastructure to the people.

Compared to the neighboring countries, vehicle ownership in Cambodia is low. There are 6.3 million motorized vehicles in the Kingdom, of which more than 85% are motorcycles. That leaves a fleet of 945,000 cars in 2020 (ESCAP, 2023).

The current accounts will have a total of zero for the base year since the number of EVs and electrical motorcycles was not found, leading us to assume that its percentage is negligible for the electricity mix now.

3.4.5 Power Capacity and Energy Yield (Current Accounts)

The installed capacity and energy generation data utilized correspond to the 2021 report from the Electricity Authority of Cambodia, (Aquarii, 2022). The distribution of shares, technological capacities, and energy generation figures for the base year 2021 are presented below. Additionally, the table encompasses information pertaining to Cambodia's electricity imports.

Table 4: Shares and values of Installed Capacity and energy yield per Fuel in 2021. (Aquarii, 2022)

Power Sources	Installed Capacity [MW]	%	Energy [GWh]	%
Domestic Generation				
Non-Renewable	1293,95	42,7%	4519,12	48,8%
Coal	675	22,3%	3813,07	41,2%
Fuel Oil	618,95	20,4%	706,05	7,6%
Renewable Energy	1738,27	57,3%	4736,62	51,2%
Hydro Power	1331,7	43,9%	4088,6	44,2%
Solar Power	376	12,4%	589,04	6,4%
Biomass Power	30,57	1,0%	58,98	0,6%
Total Domestic Generation	3032,22	75,6%	9255,74	73,5%
Import Power Sources				
Thailand	277,3	28,27%	305,68	31,17%
Vietnam	332,45	33,90%	1135,72	115,80%
Laos	371	37,83%	1903,96	194,13%
Total imports	980,75	24,4%	3345,36	26,5%
Total Power sources	4012,97	100%	12601,1	100%

3.4.6 Load Curve

The 2018 annual report presents a daily load curve depicting the typical pattern of power demand daily (Figure 11). Given Cambodia's geographical proximity to the equator, the electricity requirements remain relatively consistent throughout the year, unaffected by varying weather conditions. (Electricité du Cambodge, 2018)

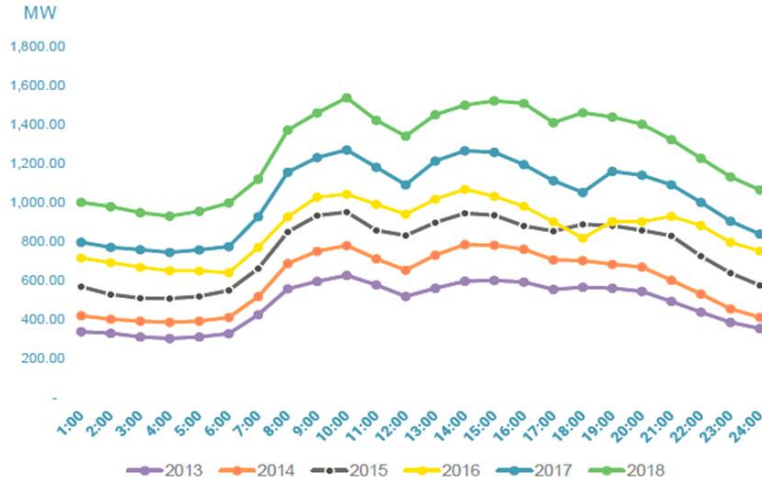


Figure 11: Daily Peak Load Curve from 2013 to 2018 in National Grid. (Electricité du Cambodge, 2018)

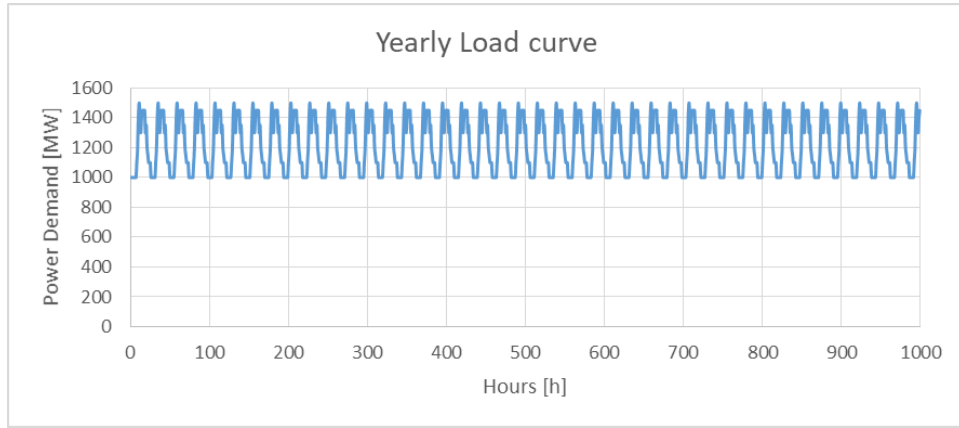


Figure 12: Yearly pattern for the load curve.

The annual load curve underwent reorganization and normalization based on peak demand, ensuring that the power demand for each hour was replicated 365 times throughout the year. To generate an input load curve for Leap, the total annual hours were segmented into 24 slices, each comprising 365 hours, and were assigned the specific power demand for each punctual hour. This results in Table 5 which is entered in the model resulting in the curve seen in Figure 13.

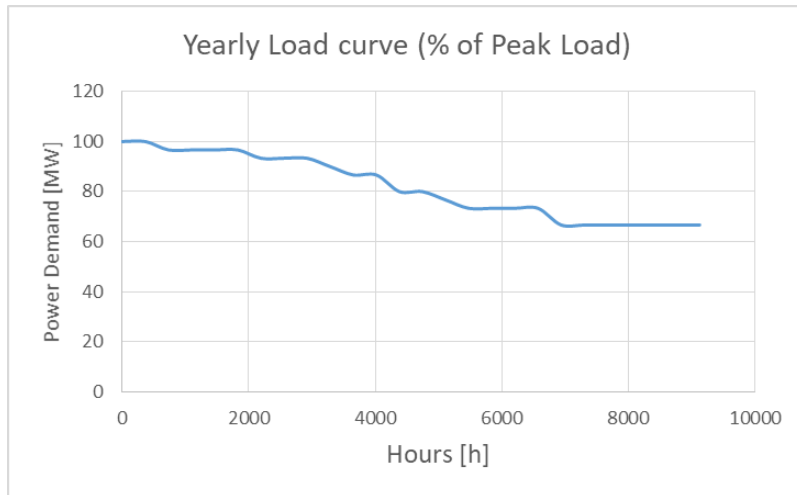


Figure 13: Yearly pattern for the load curve.

Table 5: Time slices over the year for loads over a day. Each hourly load is repeated 365 times over a year.

Day Hour	Demand (MW)	Demand/Demand Peak	Yearly hours
00:00	1000	66,7%	365
01:00	1000	66,7%	365
02:00	1000	66,7%	365
03:00	1000	66,7%	365
04:00	1000	66,7%	365
05:00	1000	66,7%	365
06:00	1000	66,7%	365
07:00	1100	73,3%	365
08:00	1200	80,0%	365
09:00	1400	93,3%	365
10:00	1500	100,0%	365
11:00	1400	93,3%	365
12:00	1300	86,7%	365
13:00	1400	93,3%	365
14:00	1450	96,7%	365
15:00	1450	96,7%	365
16:00	1450	96,7%	365
17:00	1450	96,7%	365
18:00	1300	86,7%	365
19:00	1350	90,0%	365
20:00	1200	80,0%	365
21:00	1150	76,7%	365
22:00	1100	73,3%	365
23:00	1100	73,3%	365
00:00	1100	73,3%	365

3.4.7 Transmission and Distribution (Baseline)

In 2021, 10% of the total national power supply is transmission and distribution losses, and Cambodia plans to reduce the transmission and distribution losses under 8% by 2030 (Cambodia Basic Energy Plan, 2019).

3.4.8 Technology and Costs

Various technologies and their properties (lifetime, net capacity, cost, fuel cost) were taken from the IEA projected cost of generating electricity (International Energy Agency, 2020). All the costs are assumed to be discounted at 10%. Efficiency, Availability and Capacity credit is taken from (Kamia Handayani, 2022)

Table 6: Costs and Technology Characteristics. (Kamia Handayani, 2022)

Technology	Lifetime (yrs)	Net Capacity (MWe)	Efficiency (%)	Availability (%)	Capacity Credit (%)	Capital Cost (USD/KWe)	Fixed OM Cost (USD/Mwe)	Variable O&M Cost (\$/MWh)	Decom Cost (USD/MWe)	Fuel Cost (USD/MWh)	Reference Country
USC Coal	40	347	45	80	100	974	14,97	3.71	0,01	28,02	China
NGCC	30	475	58	85	100	648	13,49	3.67	0,01	53,54	China
Fuel Oil	30		45	80	100	800	8	3	6,4		
Hydro	50	175	100	36	51	2639	11,65	6	0,001		India
Solar	25	20	100	17.7	22	766	0,16	0	0	0	China
Wind Onshore	25	50	100	28	35	1231	13,18	0	0,44	0	China
Biomass	25	30	31	80	100	2100	20	3	0,01	106	India

3.5. Demand Projections

3.5.1 Demand Households (Baseline)

The household energy consumption projections are made considering the different categories which this sector is divided into. For instance, the rate of increase for cooling might be too high compared to other categories, this breakdown makes it easier to apply specific efficiency policies.

- *Comfort*

The projection for comfort was made by following the Cambodia's National Cooling Action Plan. The values projected for Building space cooling in 2030 and 2040 have a potential of energy savings by applying according to Table 7.

Table 7: Saving potential due to improved space cooling efficiency. (Ministry of Environment , 2022)

Savings potential		
2030	2040	2050
21%	27%	31%

By using the total energy demand per year by the household sector and the projection for the whole cooling sector we calculate the share of the energy for cooling that households are demanding. We assume that this share is kept along the years until 2050.

Table 8: Total Demand Projections for the Residential sector with efficiency goals from Cooling Action Plan.

Category	Share	Base Year	Intervention with energy efficiency		
		2021	2030	2040	2050
Total Energy Demand [GWh]	100%	2515	2732	3286	3840

The projections per household by applying the efficiency goals from the Cooling Action Plan are presented as follows:

Table 9: Demand projections per household with efficiency goals from Cooling Action Plan.

2020	2030	2040	2050
Energy (kWh/H)	Energy (kWh/H)	Energy (kWh/H)	Energy (kWh/H)
403,3	628,2	672,8	696,7

- *Cooking, Recreation, Lighting, and Cleaning*

According to Figure 9, we have the percentage of different uses in household electricity consumption. From (Dul, 2020), the electricity demand in Cambodia is projected to be 70540 GWh

in 2050, in which residential sector would account 35.0%. Therefore, we calculate the electricity demand of residential sector for different end-uses in 2050.

The number of households is projected to have a 1.2 annual growth rate, and the number is 5.51 million in 2050. The National Energy Efficiency Policy proposes 17% savings in the use of electricity for the residential sector by 2030, since there is no further plan after 2030, we assume that the target for improvement of efficiency will be the same as National Cooling Plan in 2040 (27%) and 2050 (31%). By applying demand reduction due to efficiency improvement (2030,17%, 2040, 27%, 2050, 31%), we get the result of future electricity demand per household. The result is as follows.

Table 10: Electricity Demand for residential sector except for the cooling part.

Total Residential Demand (GWh)	2030	2040	2050
Cooking	2876.7	4285.7	5710.3
Recreation	935.1	1393.2	1856.3
Lighting	673.5	1003.4	1336.9
Cleaning	616.9	919.0	1224.5
Per Household Demand (KWh)	2030	2040	2050
Cooking	662.8	876.4	1036.4
Recreation	215.5	284.9	336.9
Lighting	155.2	205.2	242.6
Cleaning	142.1	187.9	222.2

3.5.2 Demand Commercial (Baseline)

The Commercial Sector in Cambodia is expected to grow aligning with the country's overall development. It was challenging obtaining official projections for the sector so estimates from a relevant master's thesis were utilized (Dul, 2020). Projecting demand for 2030, 2040, and 2050, assumptions were made based on a constant 34.8% allocation of total consumption to the commercial sector (IEA, 2023), acknowledging a potential deviation from this assumption over time.

The BAU demand projections for 2050, amount to 70.54TWh (Dul, 2020) and applying the commercial share and the Nationally Determined Contributions (NDC) policies previously presented in section 2.4 we get the following results:

Table 11: Commercial Electricity Consumption in BAU and with NDC (Baseline)

	2020	2030	2040	2050
Total Electricity Consumption (TWh)	11.11	30.92	50.73	70.54
Commercial Electricity Consumption without NDC(TWh)	3.866	10.76	17.65	24.55
Commercial Electricity Consumption with the NDC policies (TWh)	-	9.14	14.4	20.96

3.5.3 Demand Industrial (Baseline)

Cambodia's industrial sector today might be perceived as weak and unstable, with the most emphasis being on agriculture and manufacturing. Garments and food processing remain the top industries, which lack competitive advantages due to the absence of sophisticated processes, circular value chains, and technology-based production. However, the Cambodian government has shown support for the country's development plans, despite the lack of actual data available on the forecasts of sustainable economic growth.

Since the industrial sector in Cambodia remains unpredictable and scattered, the actual forecasts for industry electricity consumption could not be found. We have forecasted the demand for industries in 2030, 2040, and 2050 if the 30.078% share of electricity consumption will at least remain constant, since no data pointed us in the direction of an increase. Given that the total electricity consumption in 2050 was projected to be 70.54 TWh, 30.078% of that figure would result in 13.346 TWh of industrial electricity consumption in the year 2050. Assuming a linear trend (although not entirely accurate) would give us the projected demand for industries in 2030 and 2040 as per the table below. In addition, the NDC stated a 2.3% annual reduction in electricity consumption until 2050, which has been accounted for in the final row of the table below.

Table 12: Industry Electricity Consumption in BAU and with NDC (Baseline)

	2020	2030	2040	2050
Total Electricity Consumption (TWh)	11.11	30.92	50.73	70.54
Industrial Share of Electricity Consumption (TWh)	3.341	5.85	9.598	13.346
Industrial Share of Electricity Consumption + 2.3% annual reduction policy (TWh)	(base year)	5.773	9.444	13.115

3.5.4. Demand Transport (Baseline)

Transport is a significant contribution to GHG emissions in Cambodia. Although the fleet of cars remains small, the quality of cars and fuels are not the best. Roughly 80% of the vehicles used in the country are second-hand vehicles imported from the neighboring countries and the petrol used is not up to international standards. There are currently 6.3 million motorized vehicles in Cambodia, 85% of which are motorcycles, which none are electric. Road transport accounts for more than 90% of the modal share for passengers and freight. Cambodia's NDC projects that the number of vehicles in 2050 will reach 15.325 million and that 40% of cars and 70% of motorcycles will be electric by 2050, translating into 1.026 million EVs and 10.7275 million E-bikes (Planning, 2019).

Since electric consumption of EVs and E-bikes are difficult to find for the country of Cambodia, other assumptions have been made to simplify the projections. According to EVBOX, the average electricity consumption of an EV in the EU is 905.2 kWh, which we found more appropriate for our model compared to other data from the US. According to this number, our projection for EV consumption in 2050 would be 928.7352 GWh as per the table below.

Regarding e-bikes, the annual average usage of one is estimated to be 1600 km/year. A standard electric bike requires about 2.1 kWh of electricity per 100 kilometers. Multiplying these numbers by our e-bike fleets, we get an annual consumption of 360.444 GWh by 2050. Totaling these values, the transport sector will have an electricity demand of 1289.1792 GWh as per Table 13.

Table 13: Data for EVs and E-bikes in 2020 and 2050.

	2020		2050	
	#EVs	Annual Consumption (GWh)	#EVs	Annual Consumption (GWh)
EV	0	0	1.026	928.7352
E-bikes	0	0	10.7275	360.444
Total	-	0	-	1289.1792

3.5.5. Demand (modeling)

Finally, after all the projections for demand are inputted into the model we get the following results seen in Table 14 and Figure 14.

Table 14: Electricity Demand by Sector.

Branch	2030	2040	2050
Residential	7 752,7	10 785,2	13 829,9
Commercial	9 140,0	14 400,0	20 960,0
Industry	5 773,0	9 444,0	13 116,0
Transport	231,7	666,2	1 286,9
Total	22 897,4	35 295,4	49 192,8

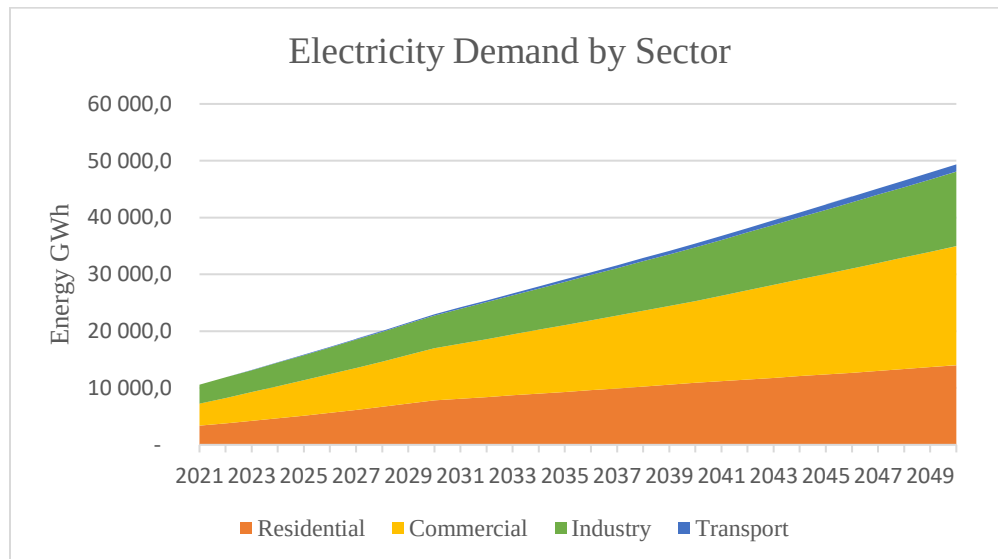


Figure 14: Electricity Demand by Sector.

Projections reaching around 25000 GWh by 2040, 35000 GWh by 2040 close to 50000 GWh by 2050, with the following shares:

Table 15: Share of electricity demand by sector.

Branch	2030	2040	2050
Residential	34%	31%	28%
Commercial	40%	41%	43%
Industry	25%	27%	27%
Transport	1%	2%	3%

Commercial becomes the sector with the highest share, while the transport sector has an almost neglecting electricity demand despite policies to increase the electrification in the sector.

The main contributor in each sector is the cooling needs. However, in the residential sector the policies of efficiency applied to the cooling sector (Ministry of Environment , 2022) makes cooking to become the activity with higher share within households.

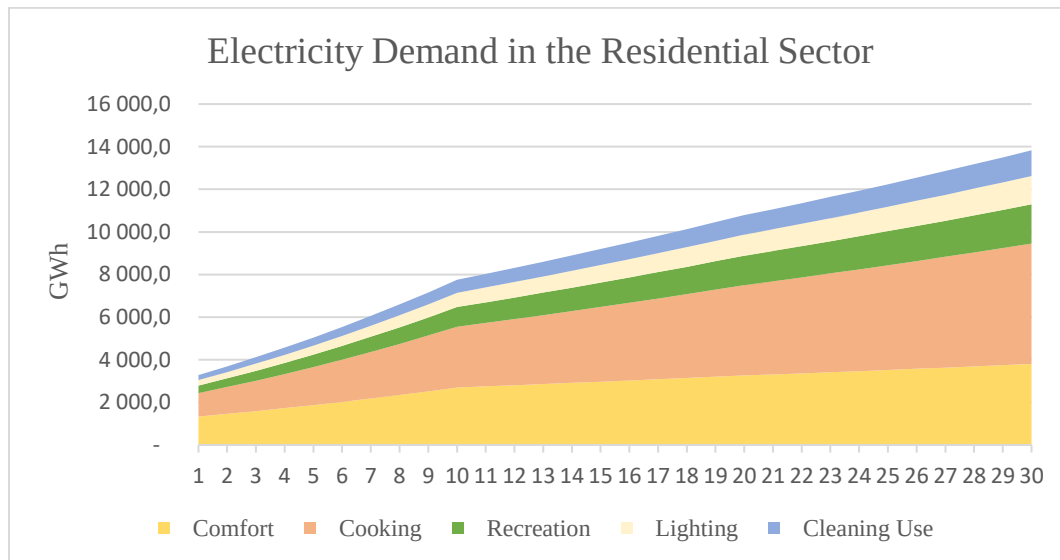


Figure 15: Electricity Demand in the Residential Sector.

3.5.6. Installed Capacity (Baseline)

The forecasted installed capacity figures are derived from the Power Development Master Plan of Cambodia 2022-2040, a collaborative effort between the Cambodian government and the Asian Development Bank. The scenario projections encompass a diversified energy mix, incorporating Coal, Hydro, Renewables, Gas-Fired Natural power plants, and the National Energy Efficiency Plan. The outlined projections extend to the years 2030 and 2040, and include estimates for energy imports, as depicted in Table 1 in Section 2.3. (Ministry of Mines and Energy, 2022)

Hydro:

Cambodia has a vast water resource which makes hydroelectricity the major source of electricity. The country is planning to build more Hydro power plants in the decade of 2020 adding 420 MW between 2022 and 2023. (Aquarii, 2022)

Sambor Hydro dam is projected to be finished by 2026 adding 600 MW and the second phase by 2027 with 1200 MW more. The power master plan scenarios show this increase in capacity for the decade of 2030. For our Leap Model we are adding this capacity in the earlies 2030s (around 2032 and 2034). With this the capacity shown in Table 13 for the projected years. (Aquarii, 2022)

Solar:

The information for solar is based on the power Master plan. Our LEAP model will progressively add solar capacity starting from the capacity in 2021 (376,8 MW) and reach 3155 MW by 2040.

Bioenergy:

Players in the country are looking for alternatives for coal such as burning biomass, pellets, wood pellets, or rice fuel pellets. (Asian Power, 2023) The value used for Leap model corresponds to the one in the Power Mester plan which can be seen in Table 13.

Coal:

Coal's share in the energy mix was set to increase to 75 percent by 2030. The Kingdom's coal power capacity is projected to reach 4,675 megawatts (MW) within the next 10 years. Of that, only 675 MW is operational, and over 900 MW is under construction. The remaining 3,100 MW, two-thirds of the planned capacity, are not clearly on track, (Aquarii, 2022)

Even though the capacity is projected to 4675 MW the power master plan uses 2266 MW as the maximum capacity reached by 2040. Many of the projects are just proposed and NDC proposes Net zero emissions for 2050. Then the only additions planned (used in our model) are the 900 hundred that are under construction and the remainder 691 MW to reach 2266MW by 2040.

Energy Leader is a company involved in the energy sector of Cambodia and currently manages two coal plants in Cambodia. According to them the 2 coal plants will be retired by 2043. (Asian Power, 2023) With this information we are taking away 675 MW of coal by 2043.

Natural Gas

In October 2020, Chinese investment in the power industry includes the 700MW coal-fired power project in Sihanoukville. However, natural gas has as principal rival solar capacity development aligned with the country's intention to net zero by 2050. (White and Case, 2021)

The master power plan includes a rise in Natural gas capacity of 900 by 2040 which we take in our model due to the fact that China has signed a contract for it.

3.6. Own Scenario

In an attempt to answer our research question and analyze the possibility of a electricity mix without fossil fuels and more renewable energy, our own scenario proposes to replace most of the fossil fuels in the electricity sector by 2050 in Cambodia by phasing out Coal in the electricity sector by 2043 and replacing it mainly with renewable sources, by maxing out their potential accounting for the country's limitations, while also not adding the planned capacity for both Coal and Natural Gas. This scenario also looks to decrease the dependency on imports in Cambodia by reducing the planned imports by 50% in 2030.

The country's renewable energy potentials are:

Table 16: Renewable Energy Resources Potential (IRENA, 2022) (Khmer Times, 2023)

<i>Renewable Energy Resources (MW)</i>	<i>PV</i>	<i>Onshore Wind</i>	<i>Offshore Wind</i>	<i>Biomass</i>	<i>Hydro</i>
	1597000	2500	88800	2500	10000

This scenario will be built inheriting the characteristics of the Baseline Scenario, creating this way a more robust and ambitious scenario than the NDC.

3.6.1 Demand (Own Scenario)

As our scenario focuses on alterations in the transformation aspect of the electricity sector, the demand side remains consistent across various sectors. This approach enhances our ability to address the initial research question, as maintaining a uniform demand allows for a clearer assessment of the feasibility of substituting fossil fuels with renewable energy sources.

3.6.2 Transmission and Distribution (Own Scenario)

Same as the demand the transmission losses remain the same as in the baseline scenario (2021, 10% of the total national power supply is transmission and distribution losses, and a reduction of losses to 8% by 2030 (Cambodia Basic Energy Plan, 2019).

3.6.3. Installed Capacity (Own Scenario)

For the installed capacity the plans related to renewables in the first scenario NDCs are kept. On the other hand, projected coal (1591 MW) and natural (900 MW) gas are decided to not be added, instead, new solar, wind and biomass are added through endogenous capacity.

Oil fuel and coal plants are decommissioned by 2030 and 2043 respectively. The renewables to be added are limited according to the potential mentioned in Table 17.

3.6.4. Energy Commodity Costs (Both Scenarios)

The global fossil fuel market is highly volatile but a general trend of prices increasing year over year is observed (Administration, 2023) (Administration, 2020).

The prices for the LEAP Simulation are considered between 2015 to 2050, but to make the calculation easier, the prices from 2021 are taken to be same as 2015 to avoid influence of recent events. In 2015, the price of Coal, Oil, and Natural Gas were reported to be 49\$/ton of coal equivalent, 52.6\$/barrel, and 2.61\$/MBtu respectively.

Determining the cost of biomass was difficult as the feedstock comes from a variety of sources like landfill gas or combustible waste but an assumption of 0.053\$/KWh is made (Dek Vimean pheakdey, 2023). It is to be noted that the study is limited to current waste generation in Phnom Phen municipality and may not scale up for the demands of the entire country but the assumption that the price remains constant through 2050 is made in the LEAP model to simplify the modelling.

The uncertainty and increasing prices in energy commodities would affect the future electricity prices of the nation whose power generation system heavily depends on fossil fuels. This could be a potential obstacle for a developing country like Cambodia and strongly incentivizes to move away from fossil fuels.

Table 15: Weighted Average Cost of fossil fuels for the Electric Power industry. (Administration, 2020; Dul, 2020)

Year	Coal	Petroleum	Natural Gas	Biomass
	Cost (\$/tce)	Cost (\$/barrel)	Cost (\$/MMBtu)	Cost (\$/KWh)
2015	49	50.8	2.61	0.053
2050	64	70	10	0.053

4. Results and Discussion

In this section, we will present and discuss the results within the context of Electricity-Cost-CO₂, but also social impact. Both scenarios will undergo individual analysis, with an additional comparative assessment.

4.1. Baseline

4.1.1 Capacity and Electricity Generation

This graph shows all the planned installed capacities for 2040 proposed by the Power Development Masterplan. We can see an increase in all the various energy technologies from 2021 to 2050. These increases match the increase in demand that is projected for Cambodia in this report.

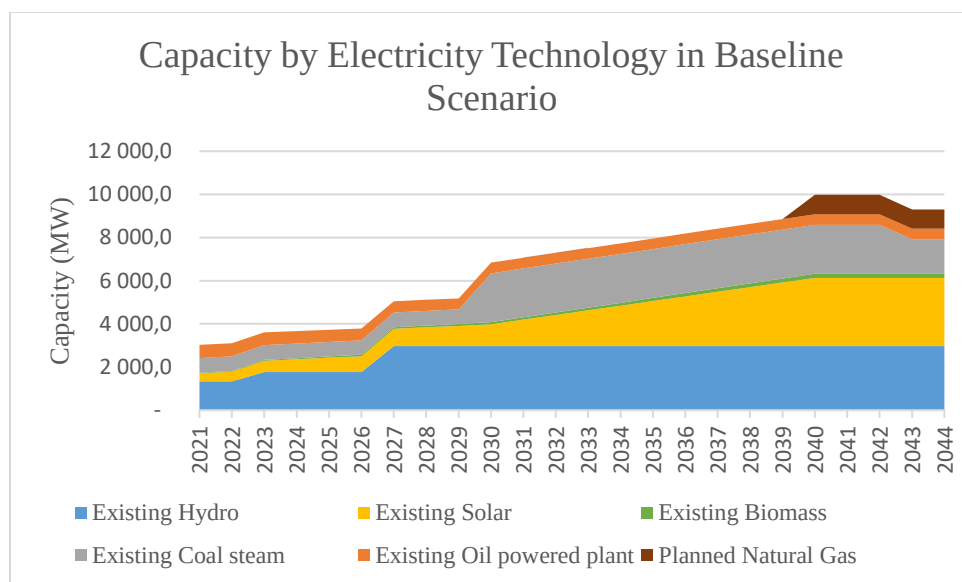


Figure 16: Installed Capacity by Energy Technology in the Baseline Scenario.

Main expansions during the 2020 decade occur in hydroelectricity as two projects under construction are scheduled to start operation, between 2026 and 2030. The decade of 2030 adds new capacity with new coal plants and progressive installation of solar capacity which now has the biggest installed capacity out of all the electricity technologies with a share of around 34% by 2050. (Refer to section 3.5.5)

Capacity is expanded according to NDC until 2040. After this year no new capacity is installed in the baseline modeling, and due to the end of lifetime, some of the power coal plants are shut down in 2042 which leads to the decrease in capacity observed.

These no addition of capacity brings the generation of electricity with oil fuel by late 2030's and the decade of 2040.

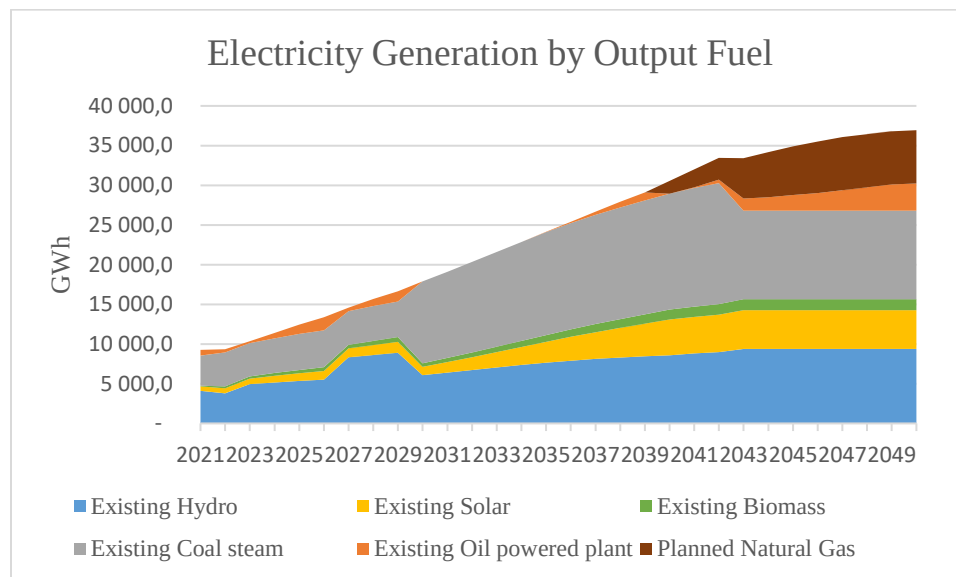


Figure 17: Electricity Generation by Output Fuel in the Baseline Scenario.

The main contributor to electricity generation is coal, followed by hydropower as seen in Table 18. Here we can see that despite the higher capacities in solar and hydro in later years these don't generate as much electricity to the mix as coal, this is due to their actual availability.

Table 17: Electricity Generation by Output Fuel in 2030, 2040 and 2050.

Branch	2030	2040	2050
Existing Hydro	6 097,9	8 600,5	9 375,7
Existing Solar	1 013,5	4 487,5	4 891,9
Existing Biomass	446,7	1 272,9	1 387,6
Existing Coal steam	10 328,4	14 567,2	11 149,7
Existing Oil powered plant	-	-	3 433,9
Planned Natural Gas	-	1 636,2	6 701,4
Total (Gigawatt-Hours)	17 886,5	30 564,3	36 940,2

The domestic energy generation does not meet the demand projections (they are around 80% along the simulation period) despite the continuous increase, so Cambodia has the necessity of importing electricity for this purpose. The figure below shows the evolution of the share of self-generated electricity and imports of electricity compared to the demand:

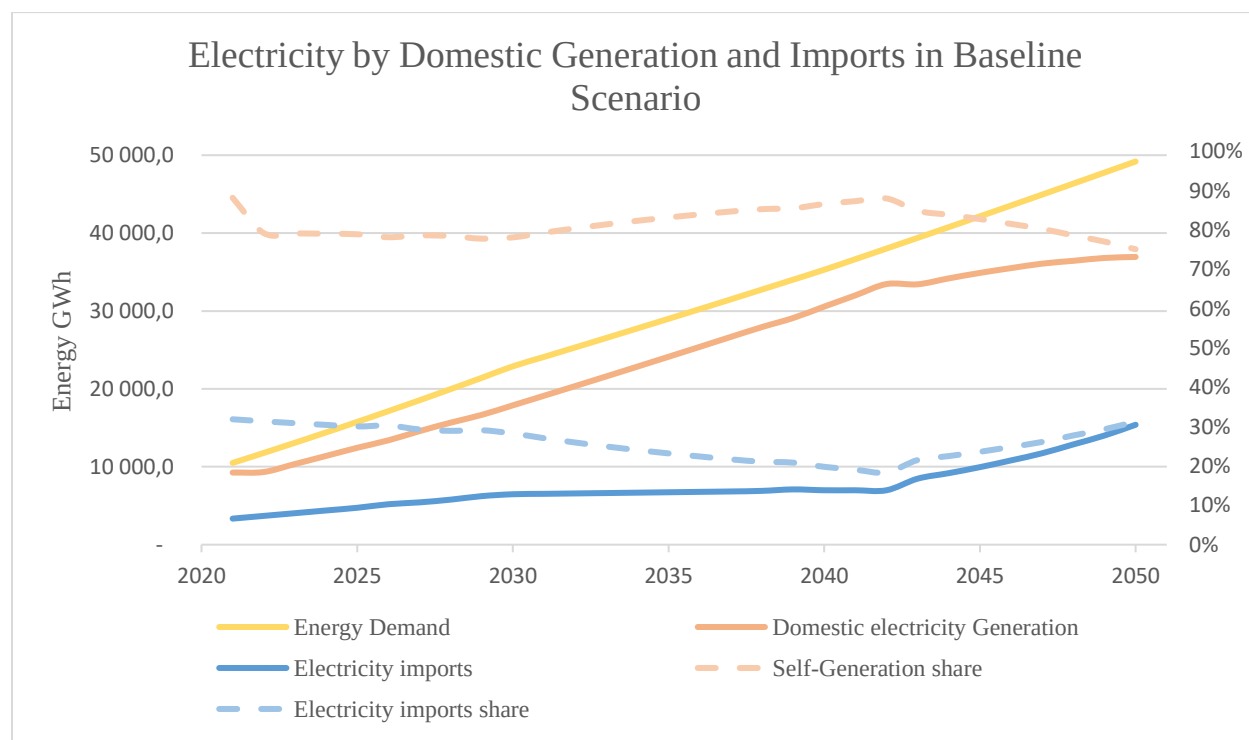


Figure 18: Electricity Demand compared to domestic and imported generation in Baseline Scenario.

In the baseline, Cambodia's electricity import will be 14 TWh per year starting from 2040 to 2050, and there will be no capacity added during that period, meaning the electricity demand will be met by the existing capacity and the import electricity.

4.1.2 Availability by source

The low availability of hydro (around 30%) and solar around (17%) brings a reduced participation of these sources in electricity generation. Even though the capacity installed for these sources is higher than coal.

By 2040 Natural gas comes into play with 900 MW of capacity installed but produces almost 17% of energy besides its lower capacity compared to Hydro or Solar.

The availability of each energy technology as mentioned in Table 6 (Section 3.4.8) highlights that renewable energy, like hydro and solar, has lower availability compared to fossil-based energy sources due to their intermittent nature. Future variations in availability are not considered but to make an accurate model it should be included if one desires to do so.

Biomass has a similar availability to coal which makes it a good candidate to replace fossil fuels. Figure 20 shows the availability of each source of energy along the simulation period:

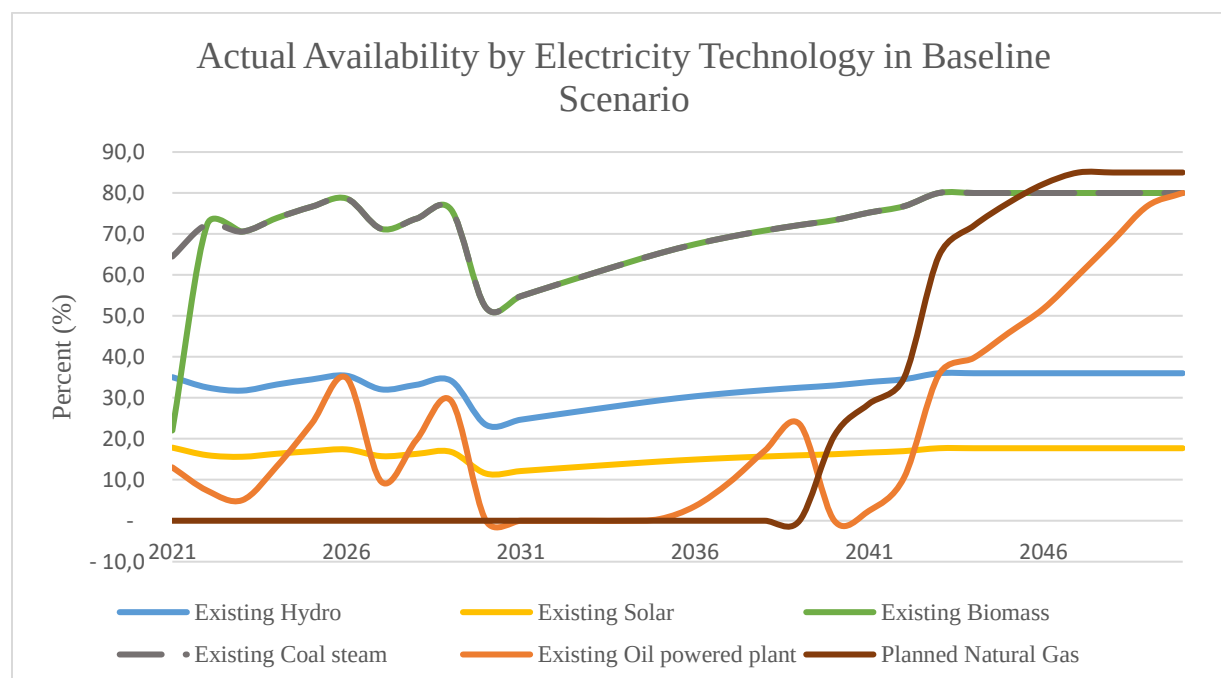


Figure 19: Actual Availability of the Energy Technologies in the Baseline Scenario.

4.1.3 Reserve Margin

Regarding the reserve margin Cambodia's plans of self-generation plus imports cover the necessary reserve margin of 20% until the year 2040. Whenever they are adding new installed capacity the reserve margin presents an increased peak and normalizes as far as demand increases.

Our baseline model does not add any new capacity after 2040 so the reserve margin starts significantly decreasing. This observation highlights the need to increase the either installed capacity for electricity generation or the imports of electricity making Cambodia more dependent in its neighbors.

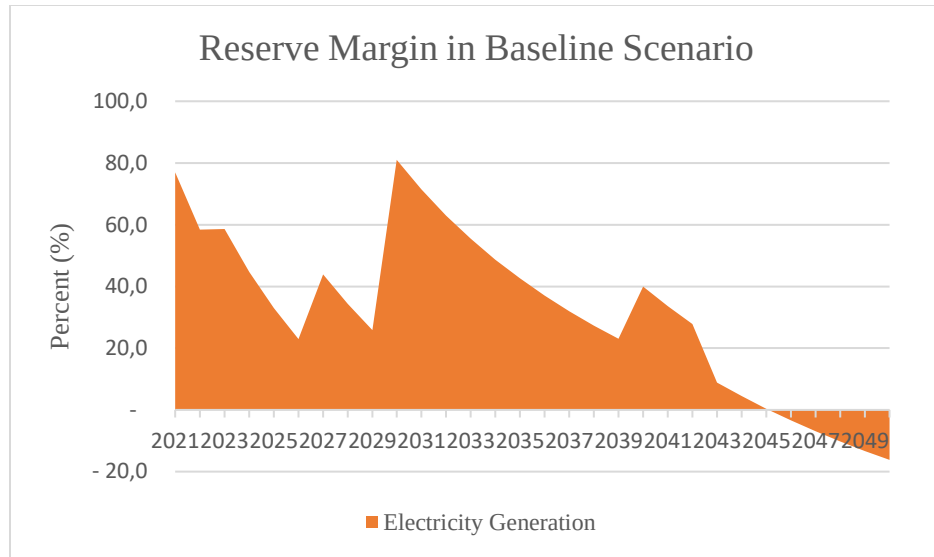


Figure 20: Electrical Reserve Margin in Baseline Scenario.

4.1.4 Emissions

Resembling the power generation trend, we can see in the figure below that there is a significant drop in emissions in the years 2023 and 2027, where new hydro capacities are planned to enter the grid.

Conversely, the Power Master Plan includes a significant addition of coal to the generation mix in 2030, placing coal as 75% of the power generation share. Coal generation capacity reaches its maximum at 2266 MW in 2042 resulting in close to 12 million tons of CO₂. However, 2 coal plants will be retired by 2050, the first one for 2043. Therefore, as per the figure below, there is a plunge in coal capacity emissions in 2043. It is visible that within the same year, there is a rise in residual fuel oil emissions, partially compensating for the sudden drop in coal capacity. Following the year 2043, the emissions continue to rise, indicating that a net zero would be highly ambitious for 2050.

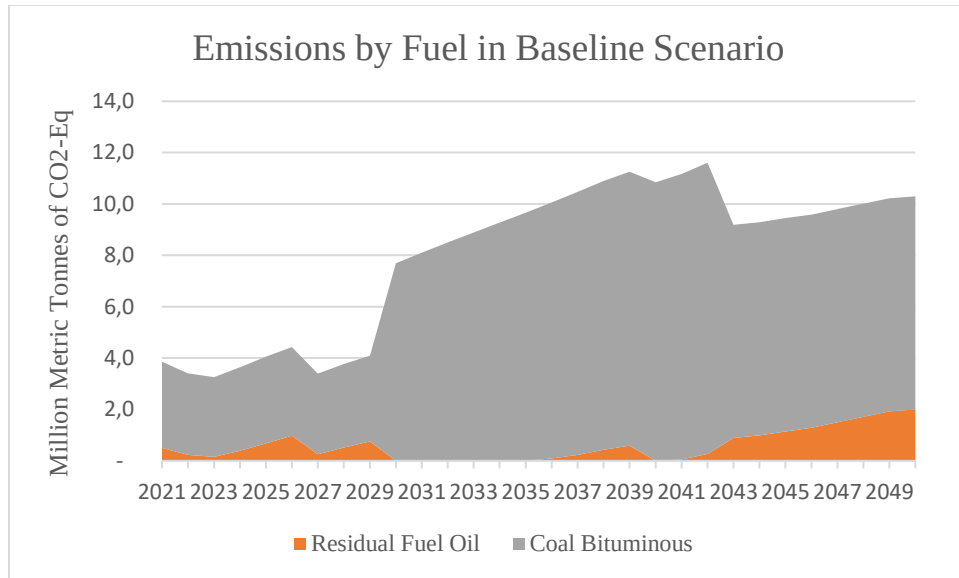


Figure 21: CO2 emissions by fuel in Baseline Scenario.

4.1.5 Costs

Social Costs in the LEAP model shows the overall social cost for the scenarios considered. These costs represent the overall costs to society of a scenario as opposed to the particular costs of production seen by producers or consumers.

Cambodia is planning to roll out carbon pricing and is in the middle of developing the policy framework (UNFCCC, 2018), we assume the price of environmental externalities to be \$9/tCO₂e to get an estimated social cost.

The overall social cost for the baseline shows that the highest cost is needed for the demand, the commercial sector has the highest social cost followed by Industrial and Residential (Figures in Annex). Insignificant cost is associated with the transport sector as the country lacks major infrastructure for energy intensive transportation. (Figure 36 in Annex). For transformation technologies (Figures 37 and 38 in Annex), we can see the associated costs for Hydropower is consistently higher compared to the rest of the technologies used. This can be attributed to the high transformation- Capital and O&M cost for the Hydropower plant.

Other technologies in the mix for the baseline scenario are the Existing Coal steam, Existing Solar, and Planned Natural Gas whose higher social cost can also be attributed to the higher Capital cost, transformation cost. We observe that the Coal and natural gas fired power plants have higher variable cost which is not significant for any other technologies considered in the baseline scenario.

For resources, Coal, Oil and Natural gas have associated social costs due to CO2 emissions associated with these fuels.

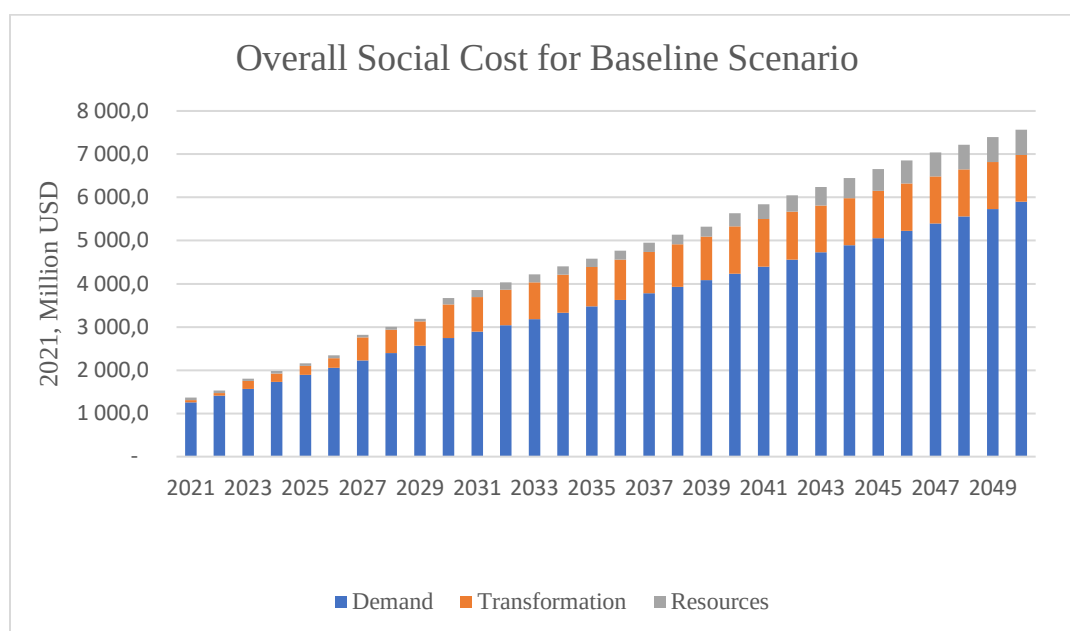


Figure 22: Overall Social Cost in Baseline Scenario.

4.2. Own Scenario

4.2.1 Installed Capacity and Electricity Generation

The model for our own scenario decommissioned oil and coal steam powered plants while capacities of renewables planned on the power master plan were added. New renewable capacities on top are left to be added through endogenous capacity according to the balance of energy regarding demand and reserve margin.

In total, in this scenario, there's more installed capacity in the country which is due to the reduction in imports we chose to achieve a more self-sustaining country (Figure 23). While in the baseline scenario, the country has around 9GW of installed capacity, in our own scenario we have 17GW. This increase is also due to the less availability of renewable energy sources compared to fossil fuel ones, which leads to a need for more installed capacity.

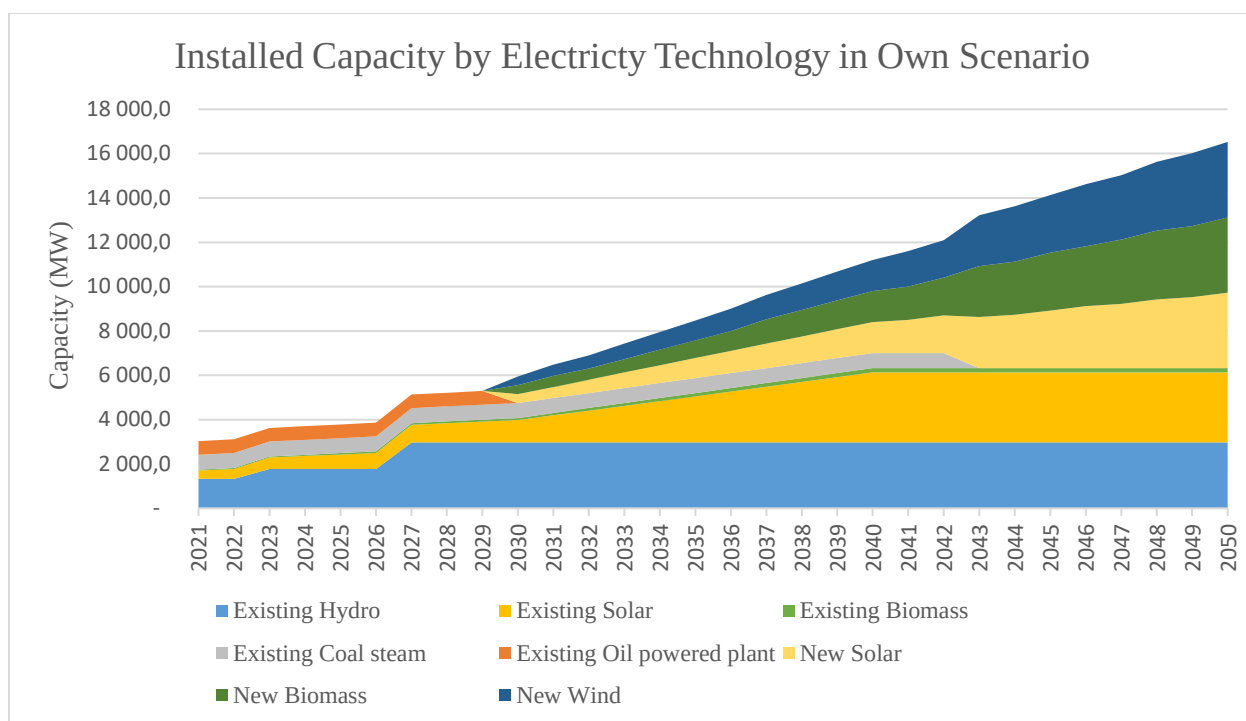


Figure 23: Installed Capacity by Electrical Technology in Our Own Scenario.

Table 18: Installed Capacities by Technology in Our Own Scenario.

Branch	2025	2030	2035	2040	2045	2050
Existing Hydro	1 773,0	2 973,0	2 973,0	2 973,0	2 973,0	2 973,0
Existing Solar	656,0	1 005,0	2 080,0	3 155,0	3 155,0	3 155,0
Existing Biomass	60,5	98,0	148,0	198,0	198,0	198,0
Existing Coal steam	675,0	675,0	675,0	675,0	-	-
Existing Oil powered plant	619,0	-	-	-	-	-
New Solar	-	400,0	900,0	1 400,0	2 600,0	3 400,0
New Biomass	-	400,0	800,0	1 400,0	2 600,0	3 400,0
New Wind	-	400,0	900,0	1 400,0	2 600,0	3 400,0
Total Capacity (MW)	3 783,5	5 951,0	8 476,0	11 201,0	14 126,0	16 526,0
Share of Renewables	66%	89%	92%	94%	100%	100%

In Table 19 we can see that biomass has more installed capacity than the theoretical maximum potential for Cambodia, this can be justified by a substantial increase in imports of biomass into the country to fulfill this installed capacity (Figure 25). The remaining capacities are within their potential, so to fully understand their feasibility a further analysis of social cost will be made in this report.

New additions are new solar, biomass and wind and with this new configuration the results imply that the country could become 100% renewable by 2045 after coal is fully phased-out of the electricity mix.

Energy generation is shown in Figure 24 as follows. The electricity generation increases through the years to match the increased demand, and, when compared to the baseline scenario we have the increase to compensate for the decreased imports from the neighboring countries, and the reduced availability of the used technologies.

Table 19: Electricity Generation by Output Fuel in Our Own Scenario.

Branch	2030	2040	2050
Existing Hydro	8 007,1	7 946,9	7 949,9
Existing Solar	1 330,8	4 146,4	4 148,0
Existing Biomass	586,5	1 176,1	1 176,6
Existing Coal steam	4 039,9	4 009,5	-
Existing Oil powered plant	-	-	-
New Solar	529,7	1 839,9	4 470,1
New Biomass	2 394,0	8 316,1	20 203,9
New Wind	837,9	2 910,6	7 071,4
Total (GWh)	17 725,9	30 345,5	45 020,0

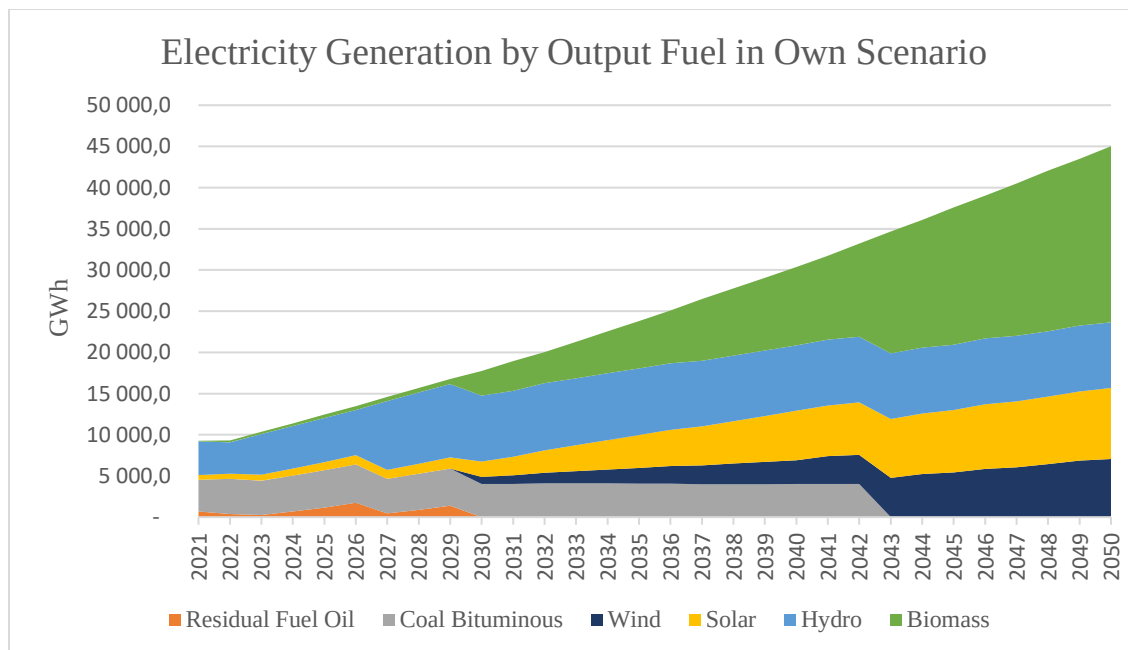


Figure 24: Electricity Generation by Output Fuel in Our Own Scenario.

On a first sight, renewables composing this mix are mainly of low availability and present uncertainties and risks regarding to its possibilities of usage as baseload of demand. The only source to which supply can be predicted and assured in advance would be biomass. Hence, additional storage solutions would need to be addressed.

Reliance on biomass might exceed the possibilities of the country regarding its biomass production bringing the need to import this resource.

In this scenario, biomass mainly covers the baseload demand of the country past 2040, this is a flexible energy resource that can be inexpensive and more easily integrated into the electricity mix of the country. Besides biomass, solar wind and hydro cover the remaining demand in mostly equal parts after the phase-out of coal in 2043. Before that, we can see that by 2030 no more oil is used in the electricity mix in Cambodia. The biomass comes mostly from imports as we can see in Figure 25.

In our own scenario, we want to decrease Cambodia's electricity import by half compared with baseline, meaning the electricity import will be 6472 GWh in 2030 and 6984 GWh in 2040, respectively. In our own scenario, Cambodia's electricity import will be 7 TWh per year starting from 2040 to 2050, but the electricity demand will not be met only by the existing capacity and the import electricity. From our result, 2510 MW of Biomass, 1250 MW of Solar PV, and 600 MW of wind will be added to the installed capacity during that period.

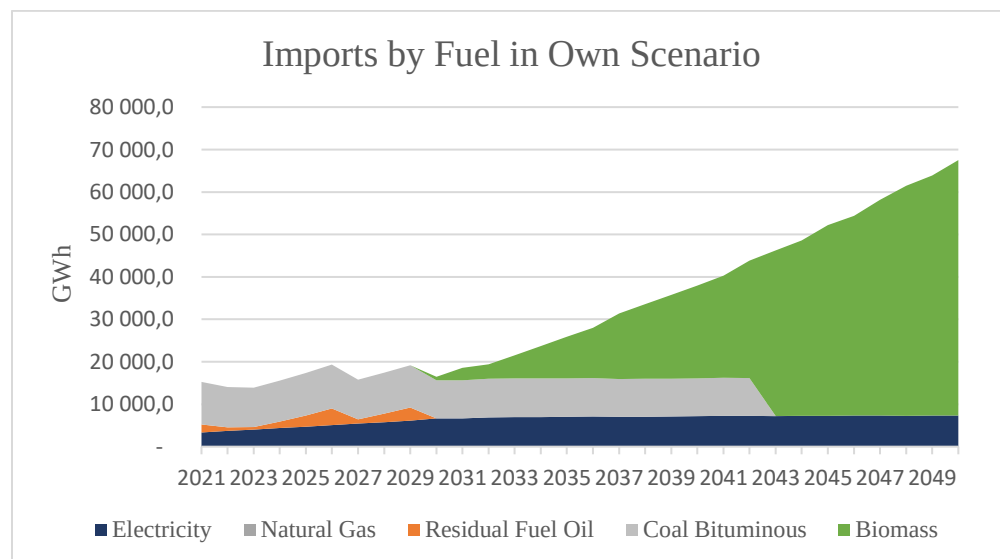


Figure 25: Imports by Fuel in Our Own Scenario.

Under this scenario, the higher capacity installed allows the domestic generation of electricity to be higher which also reduces the need for the country to import electricity. This own vision tried to explore the reduction of imports of electricity during the upcoming years by reducing it to half of the projections in the baseline.

The evolution of domestic generation and imports regarding the demand are presented in the figure:

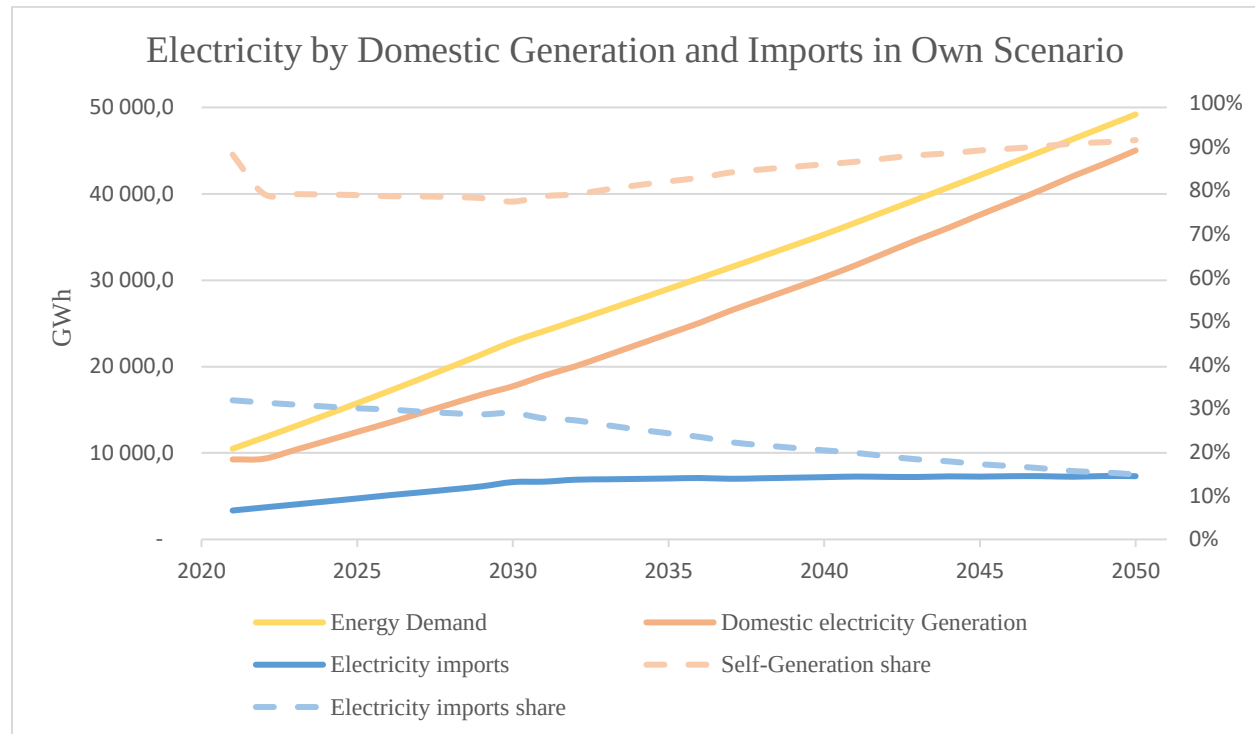


Figure 26: Electricity Demand compared to domestic and imported generation in Our Own Scenario.

Results show that with the capacity proposed the country would be able to reduce the imports from neighboring countries to less than 20% of the total demand, compared to the 30% in the baseline scenario. With this we can conclude that Cambodia has the electrical and resource potential to phase out fossil-fuels by 2050 while also reducing their imports, but costs are an important factor when applying these policies and changes to a country's electricity mix, especially in a developing country, so an analysis of social cost is going to be made to fully conclude on the feasibility of this scenario.

4.2.2 Availability

Availability is based on the entered values for the associated technologies, since the capacities for existing oil powered plants and existing coal steam are decreased as a step function and not as interpolation in the LEAP model, this results in the short-lived negative availabilities of these transformation technologies. The availabilities are unchanged from the baseline and this can be seen in Figure 27 below.

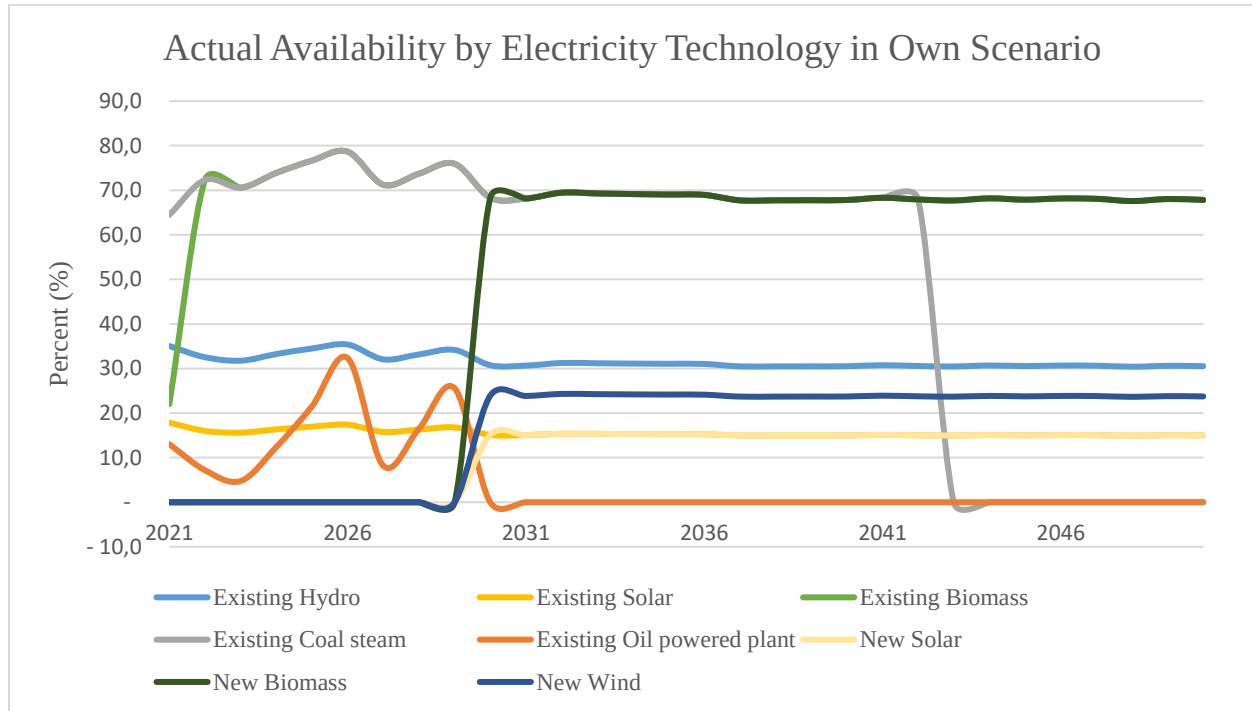


Figure 27: Actual Availability of each Electricity Technology in Our Own Scenario.

4.2.3 Reserve margin

The reserve margin presents a similar behavior during the first decade as the additions of hydro capacity are kept the same. After 2030 the model itself adds the capacity needed to keep the 20% of margin defined. The peaks related to the addition of the coal plant do not come into play.

With this feature the energy mix obtained for our own scenario assures the reserve margin.

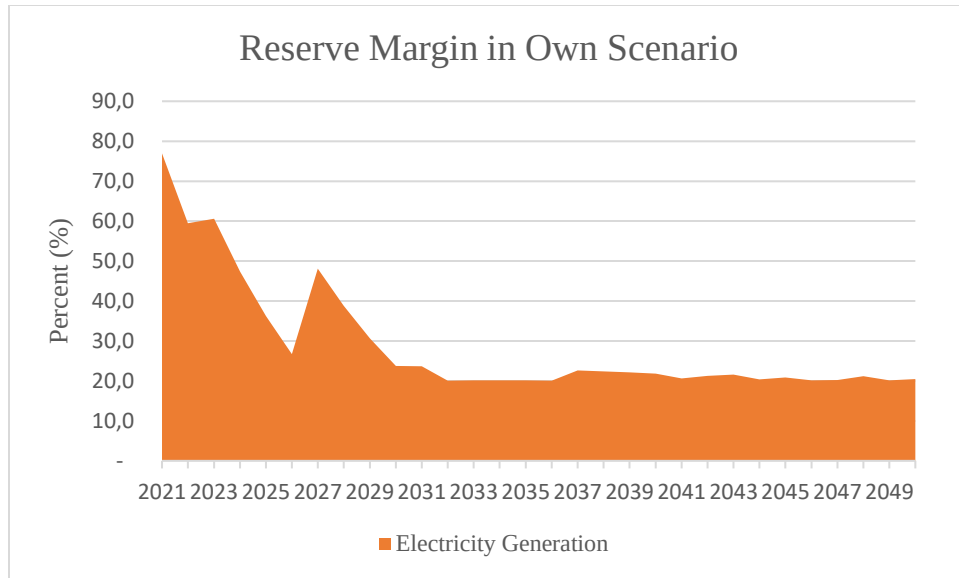


Figure 28:Electrical Reserve Margin in Our Own Scenario.

4.2.4 Emissions

In our proposed scenario, a great shift from the baseline scenario is envisioned in terms of the electricity generation capacities. Our scenario outlines the phase out of coal and fuel oil. This is in coherence with the electricity generation by output fuel graph, indicating a need for reliable base load. A significant change compared to the baseline scenario is our decision to phase out fuel oil and coal entirely in the years 2030 and 2043 respectively. Notably, the transition to renewable energy is underscored by a strategic deployment of biomass, primarily covering the baseload demand post-2040. New additions to the energy mix include capacities from solar, biomass, and wind sources.

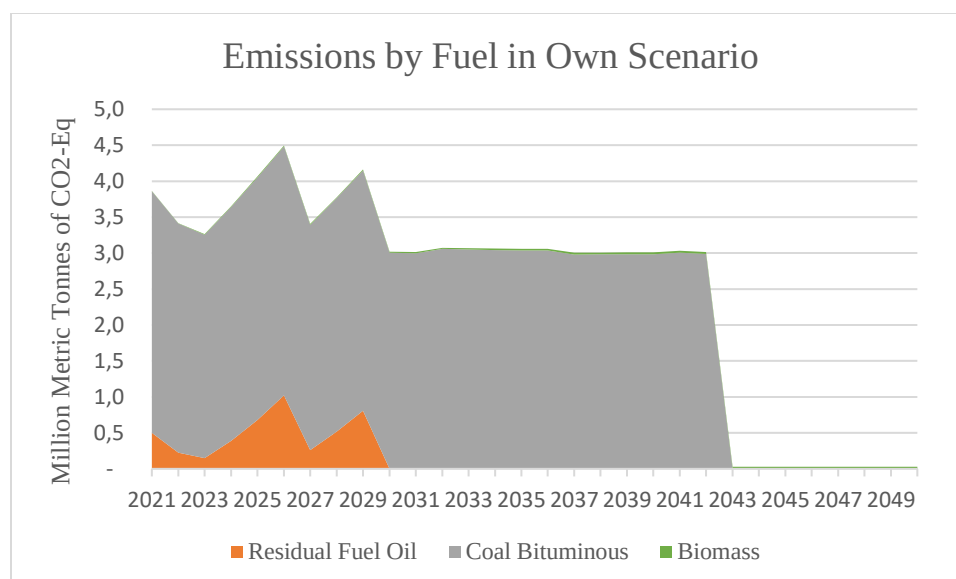


Figure 29: Emissions by Fuel in Our Own Scenario.

4.2.4 Social costs

In our scenario, the overall social cost for resources is high because of maximum utilization of biomass, the demand for biomass is extremely high and as such there is a need to import biomass to meet the energy demand which adds to the cost.

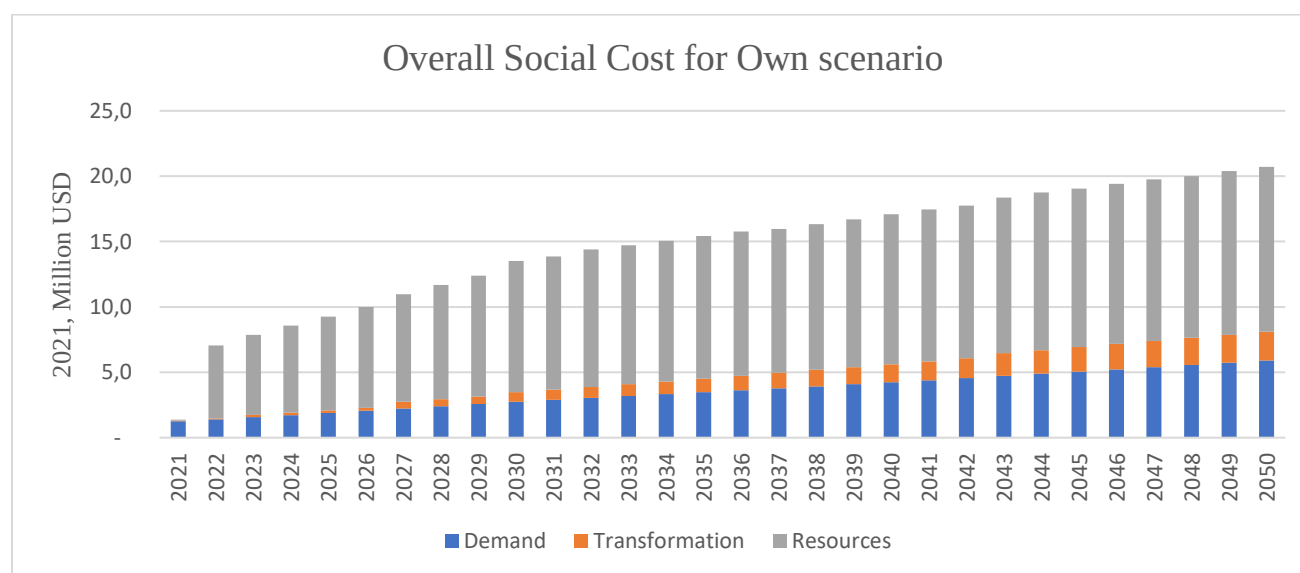


Figure 30 Overall social cost for our scenario

From the demand side, the commercial sector has the highest social cost followed by industry and residential sector (Figure 39 in Annex).

For Transformation processes, addition of new biomass (additional capacity) adds to high social cost for this process. (Figure 40 in Annex)

The high utilization of biomass needs additional imports to meet the requirement, this could affect the country's long-term goal of energy security and make it dependent on other sources of biomass for bioenergy.

One proposed solution to reduce biomass import could be to add more hydropower capacity. Additional research needs to be done to find the optimum energy mix for our low emission scenario. Additional hydropower does reduce the biomass imports, but one needs to consider the effect of climate change on river basin and FOLU emissions which are associated with biomass.

4.3. Comparison between both scenarios

4.3.1 Installed Capacity and Electricity generation

With the visual representation between both scenarios for capacity seen in Figure 31, we can see the clear reduction of coal, oil and natural gas in the electricity mix of the country, while new renewable energy resources are built to compensate for this reduction but also, to guarantee less imports and a minimum 20% reserve margin every year. New wind, biomass and solar have similar growths throughout the simulated period.

For generation (Figure 32), the trend follows the pattern seen in the installed capacity, with the decrease in fossil fuel electrical generation which is then balanced out by the increase in new renewable sources. Existing hydro and solar have a very slight decrease in our own scenario when compared to the baseline due to the installation of the new renewable sources' plants. But they remain more or less the same between both scenarios, with the majority of the differences lying in fossil fuels and new renewable energies in the electricity mix.

While for Capacity we can see a significant difference between both scenarios, when observing the generation graph, we can conclude that this difference is not as noticeable since as previously seen, the actual availability of renewable energy sources is lower than in fossil fuels which requires more capacity to fulfill the same generation. But still, there's more generation in our own scenario in order to account for the reduced imports.

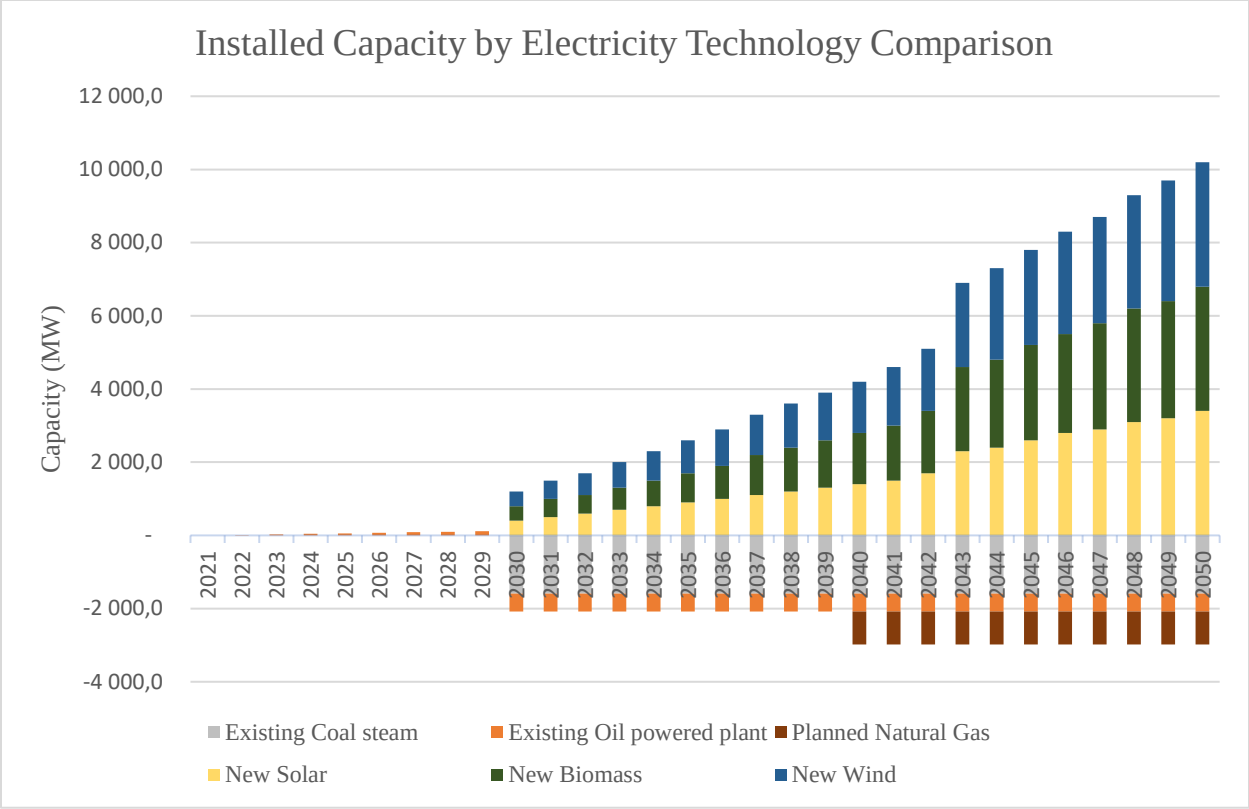


Figure 31: Installed Capacity by Electricity Technology Comparison.

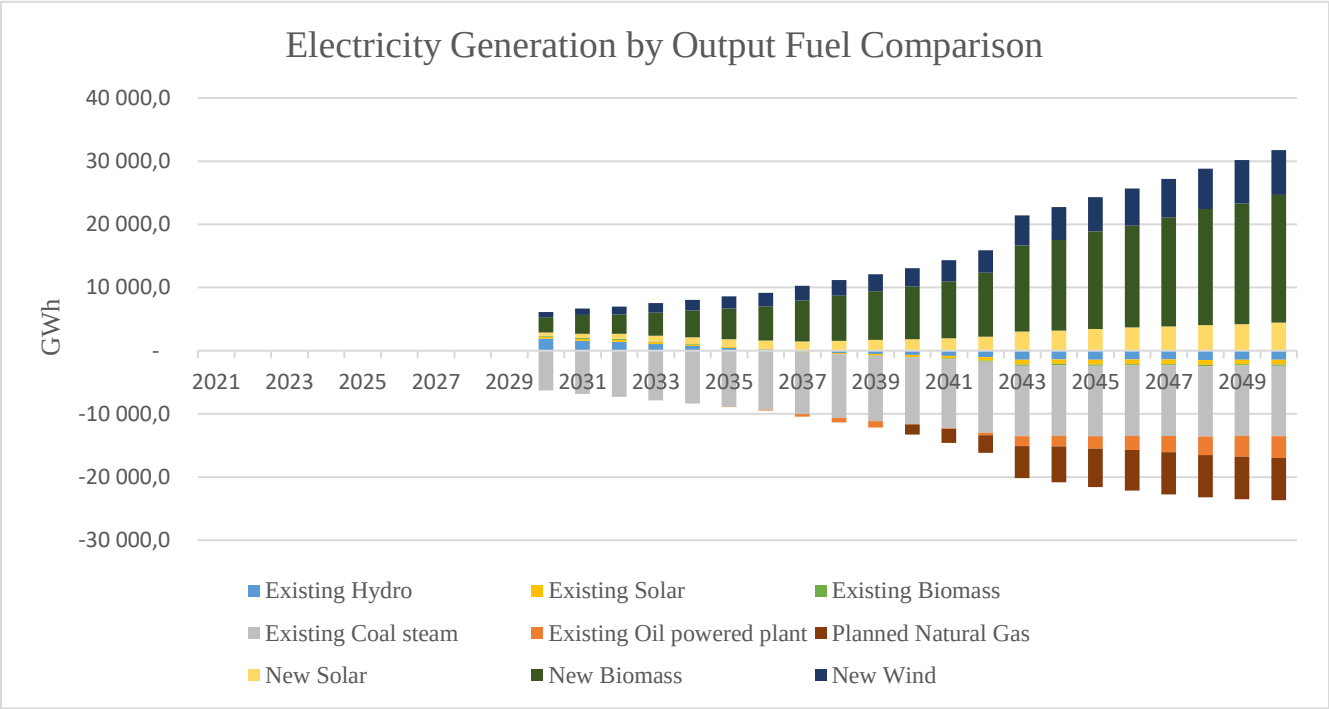


Figure 32: Electricity generation by Output Fuel, Scenario Comparison.

4.3.2 Emissions

As clearly depicted in the figure below, our proposed scenario reduces carbon emissions drastically compared to the baseline scenario. With the addition of renewable capacity into the country’s generation mix starting with hydropower capacity in 2030, our scenario presents promising results in terms of reducing the overall emissions. As evident, given our outlook, the country would edge close to carbon neutrality until 2030. Post-2030, our scenario demonstrates a steady and sustained negative emission trend, indicating the country's proactive role in sequestering more carbon than it emits. By 2050, our projections suggest that Cambodia would successfully remove over 10 million tonnes of CO₂ from the atmosphere, further solidifying its net-zero carbon footprint.

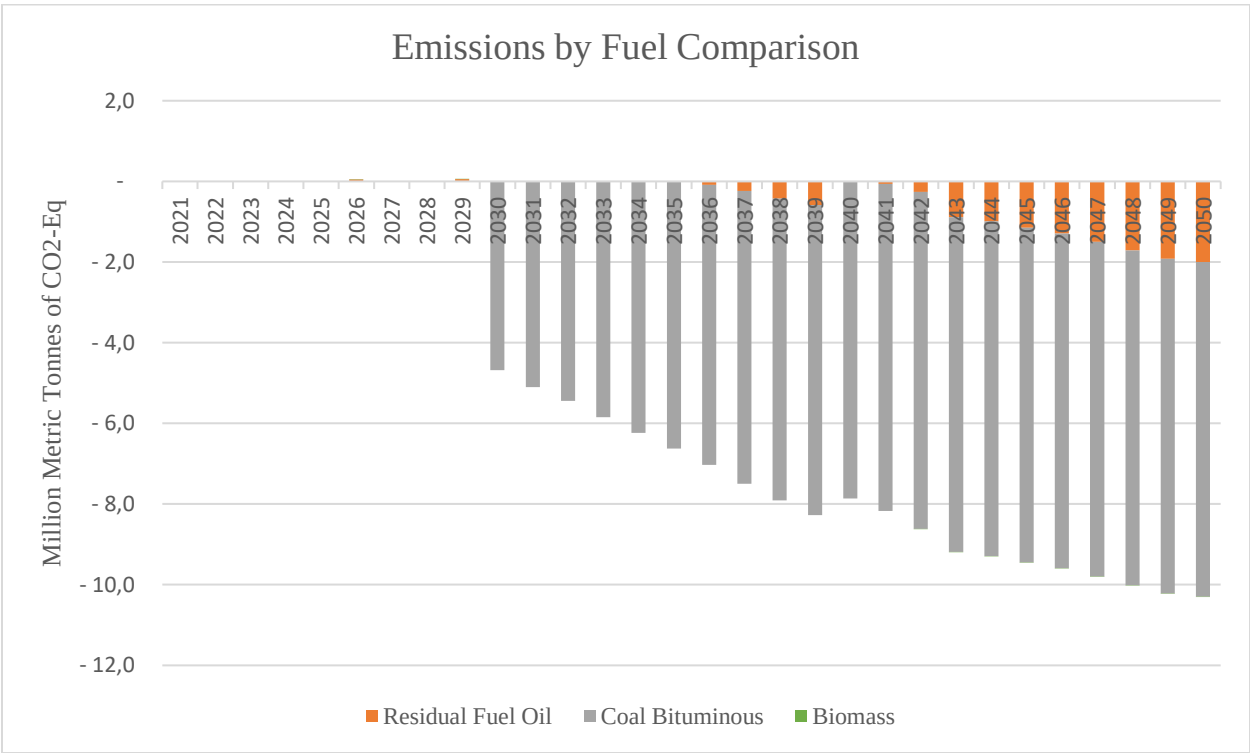


Figure 33: CO₂ Emission Comparison between both scenarios.

4.3.3 Social costs

The overall social costs for demand remain unchanged as the demand side is kept unchanged in the baseline and our scenario, however, the changes in transformation processes i.e. reduction and eventual phasing out of coal and natural gas and replacing them with renewables (majorly biomass) can be visualized in the Figure 35. A comparative analysis can be made with the GDP forecast and

social cost, for 2049 the total GDP for the country is expected to be 61.774 Billion USD and the additional social cost for the same year comes to 13.1 Billion USD. This ballooned up cost figure makes our envisioned scenario less likely. The resources associated with transformation process especially import of biomass to replace fossil fuel powered energy system needs to be fueled with increased imports of biomass whose sustainability and environmental impact needs to be considered even when the LEAP model shows reduction in greenhouse gas emissions.

Figure 34: GDP forecast for Cambodia.

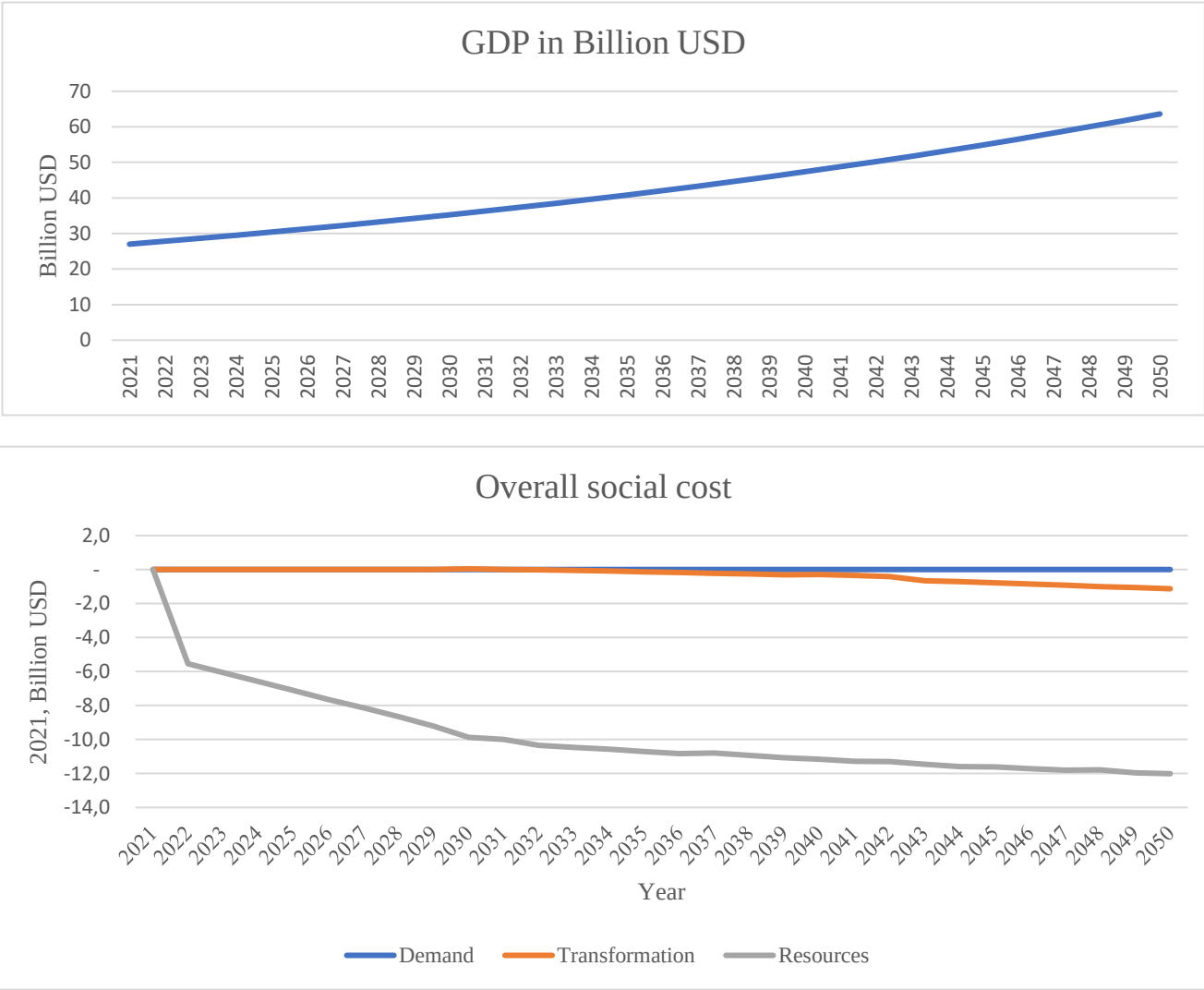


Figure 35 Difference in overall social cost from baseline scenario to our own scenario.

5. Conclusion and Policy Recommendations

It is evident that the interplay between electricity generation, cost implications, and emissions presents complex challenges and trade-offs for Cambodia's energy sector.

Within the NDC plans of Cambodia, even though many policies address various Sustainable Development Goals (SDGs), including universal access to electricity and modernizing energy infrastructure, brings conflicting aspects, particularly in relation to increasing fossil fuel capacity and its impact on overall emissions.

The own scenario explores the possibility of the potential for renewable energy sources to replace planned fossil fuel capacities, offering a pathway to meet growing demand and phase out existing fossil fuel power plants. However, it also underscores the economic considerations, revealing that the own scenario, while environmentally favorable, proved to be economically difficult compared to the NDC and other government policies.

The contrast of the two scenarios brings the necessity of a nuanced understanding of the trade-offs between environmental goals and economic development. A call for a balanced energy plan for Cambodia country acknowledges the country's right to develop its economy and underscores the importance of an effective and affordable energy plan, even if it entails increased CO₂ emissions.

From a global perspective, Cambodia seems to have the willingness on working together with neighboring countries. This collaboration might allow the country to allow addressing emissions challenges and reach net zero emissions.

5.1. Answering the Research Questions

- To what extent can the planned fossil fuel capacities be replaced with renewable energy sources?

In terms of potential combined with imports, Cambodia has the possibility to install more renewable energies to meet its growing demand and phase out its existent fossil fuel power plants without the need to install more.

But in economic terms our own scenario proved to be far more expensive than the NDC and other government policies, and in a developing, growing country that is still economically recovering from war, an expensive energy plan is not the best option. Like its already developed counterparts, Cambodia has a right to develop its economy and electricity and energy systems is an effective

and inexpensive way even if it means more CO₂ emissions for the world. In order to counterbalance, developed countries in the west should increase their efforts to decrease emissions in order to allow developing countries to continue growing in the most organically way possible.

So, while renewable energies have great potential and should continue to grow in Cambodia as years progress, the complete phase-out of fossil fuels in the electricity mix by 2050 is not the most viable option for the country any time soon.

- Are the NDC and government policies in line with sustainable development?

While a lot of the SDGs are answered through many of the policies in the NDC and other official government documents, like SDGs 1, 5, 8, 9, 10 and 13, as seen in the introduction, Cambodia is still lacking in this aspect. And when talking about SDG7, *Affordable and Clean Energy*, NDC and other policies both have policies that search to work towards this goal but others that work against it. As seen with the report, Cambodia searches to ensure universal access to electricity in the country by 2030, improve energy infrastructure and services so they are more modern, it is achieving this by increasing fossil fuel capacity in the country which increases the overall emissions counteracting against the remaining efforts towards achieving SDG7.

5.2. Policy Recommendations

- a. Promote energy efficient equipment in households for cooling, cooking and lighting like LED, high-EER air conditioners and high-COP refrigerators. Cambodia's ambitions regarding this sector are on track to net zero emissions.
- b. Promote the phase-out of coal power plants and the deployment of renewable electricity generation; the focus should be on biomass, solar, and wind.
- c. Provide subsidies to renewables: the feed-in tariff and feed-in premium should be considered to attract more investments in renewable energy. As there is potential for renewable sources such as wind and biomass which are not being considered in actual energy plans.
- e. Increase investments in power transmission and distribution infrastructures in order to further improve electrification and electricity development in the country.

f. The observations on the transport sector show that an increase in EVs will not stress the system by generating a high demand. Hence, Promote EVs and bio-fuel vehicles by providing tax subsidies will reduce the load of emissions and facilitate the reach of net zero emissions.

g. International communities should provide more financial and technical support to Cambodia to allow it to grow more sustainably and not rely on more fossil-fuels.

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Annex

Costs in Baseline Scenario:

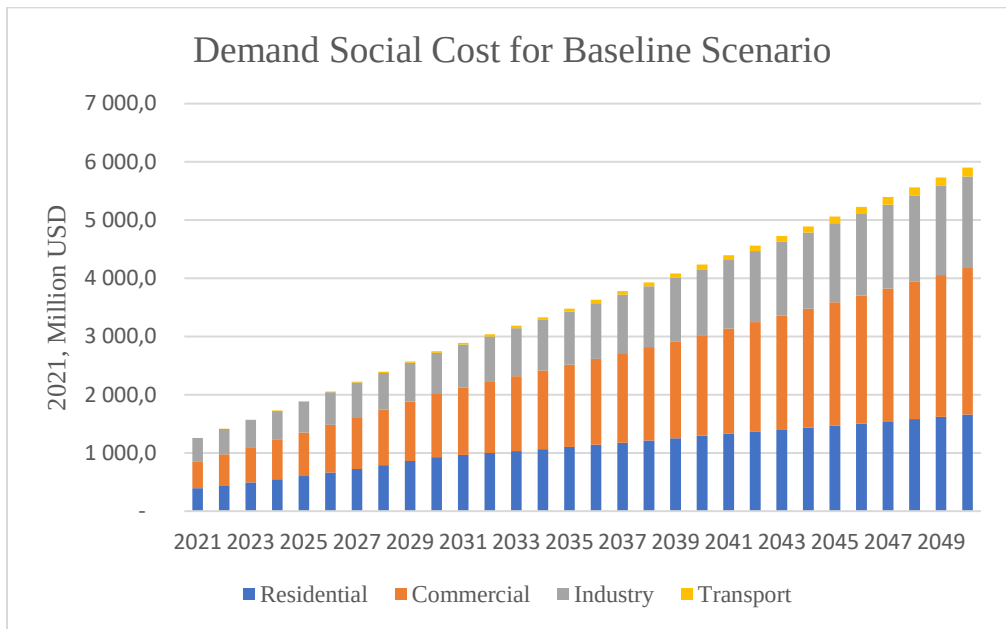


Figure 36 Demand Social Cost for Baseline Scenario

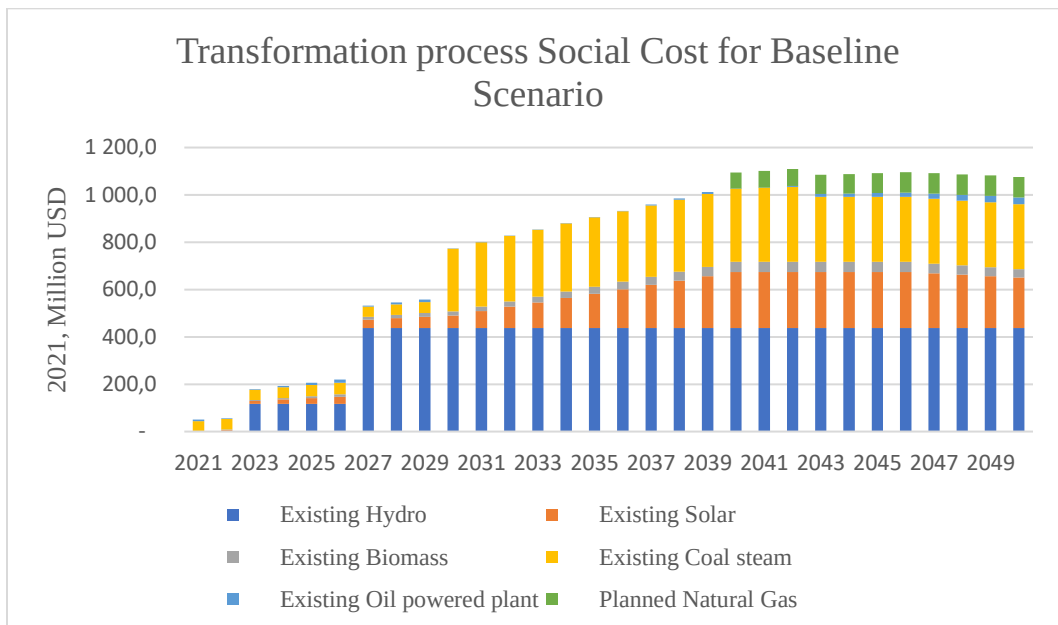


Figure 37: Transformation process Social Cost for Baseline Scenario

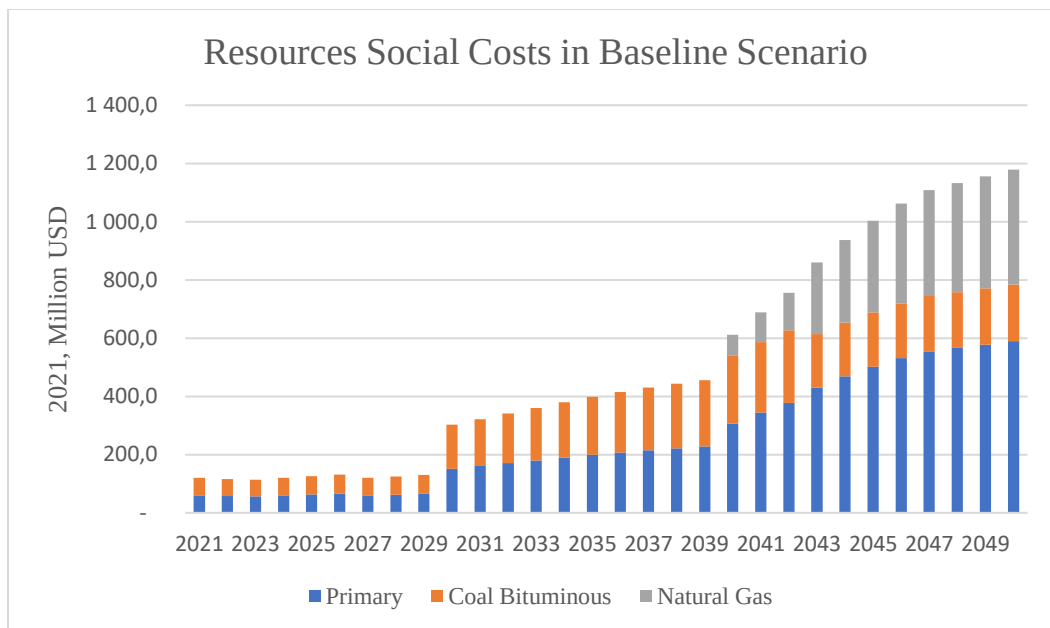


Figure 38: Resources social costs in baseline scenario.

Costs in Own Scenario:

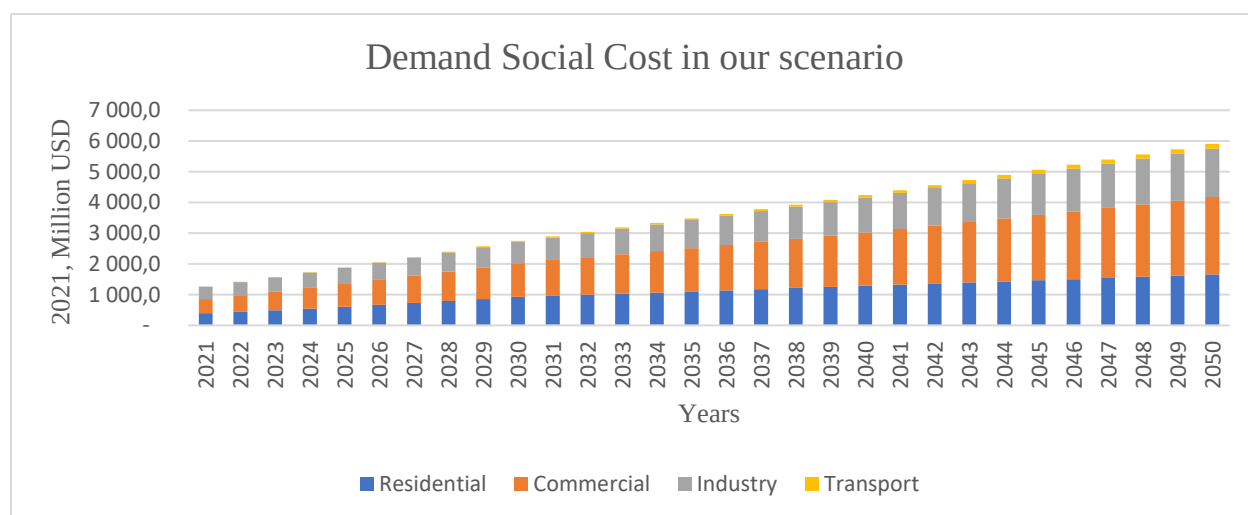


Figure 39: Demand Social Cost in our scenario.

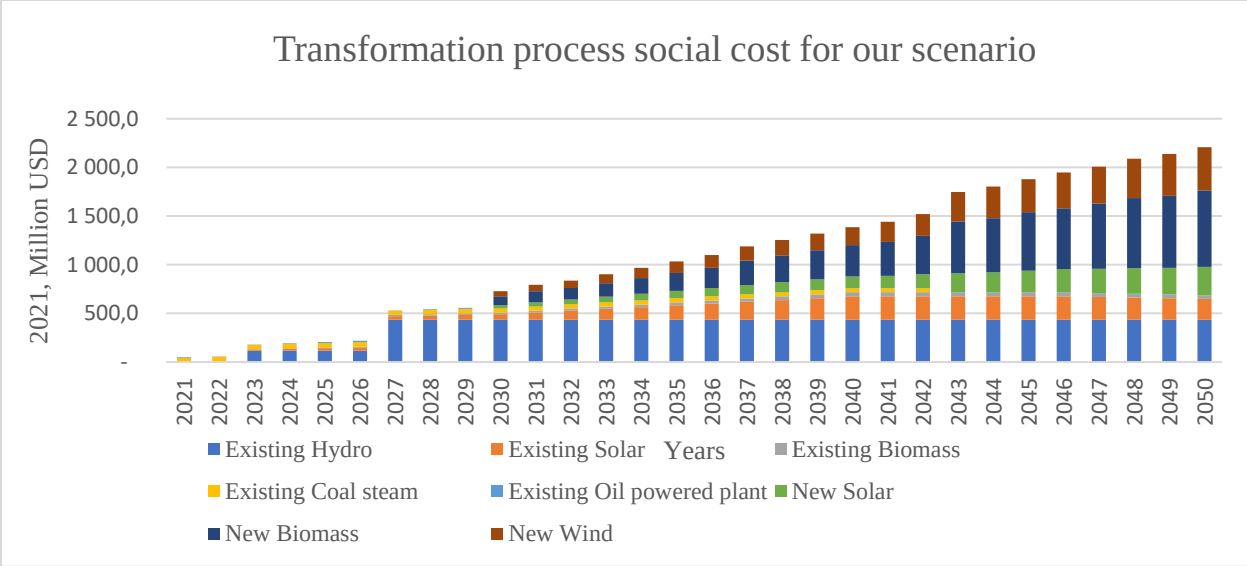


Figure 40 Transformation process social cost for our own scenario

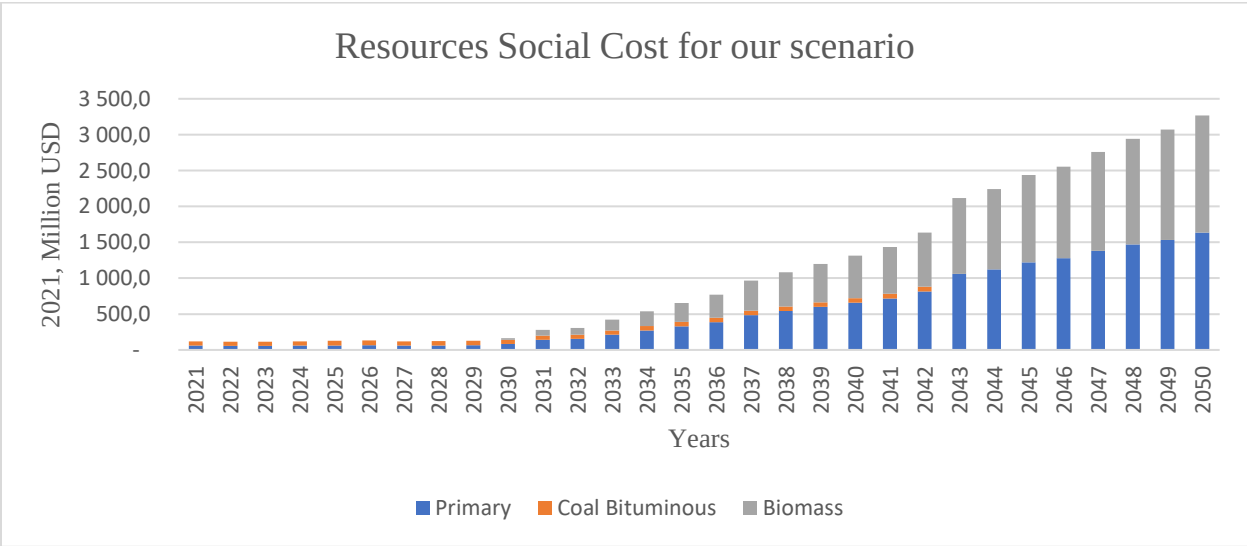


Figure 41 Resources Social Cost for our scenario

Social Cost Comparison:

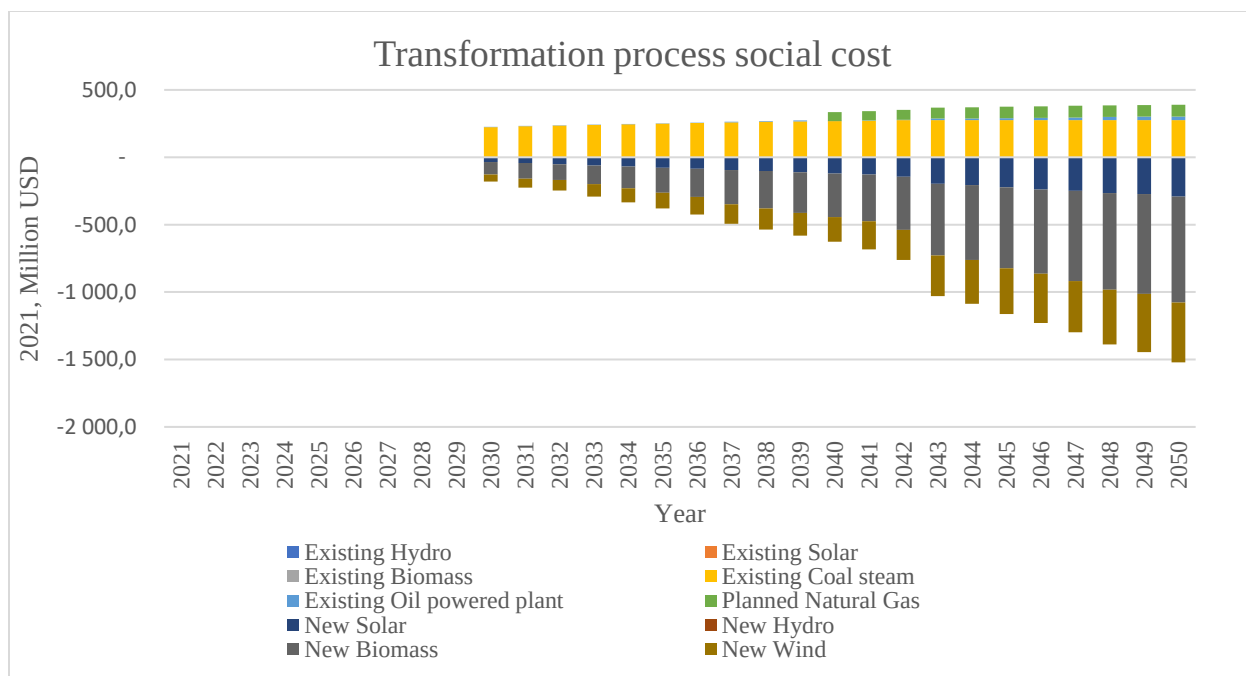


Figure 42 Difference baseline and our scenario transformation process social cost

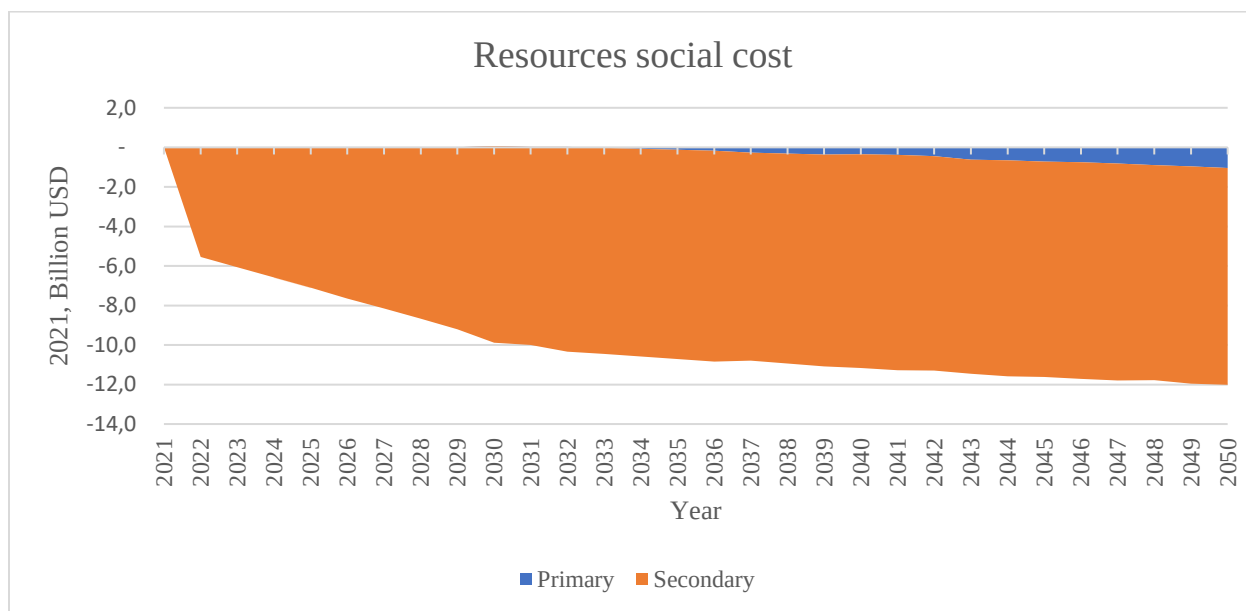


Figure : 43 Difference baseline and our scenario resources social cost